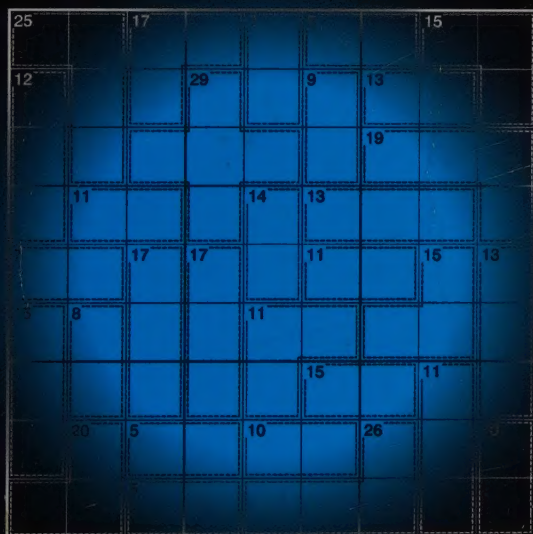


THE  TIMES

ULTIMATE KILLER BOOK 2

Su Doku



The deadliest of all Killer collections

THE  TIMES

Ultimate Killer Su Doku

Book 2

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Book 2

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Introduction

Welcome to the second edition of *The Times Ultimate Killer Su Doku*. Killers are my personal favourites, and this level is the pinnacle!

Let's start with the challenge – the top players in the world will be able to solve each of these in 20–30 minutes. To match them, the first step is to develop an efficient set of techniques that enable you to solve every puzzle. The ones described here are sufficient. As you work through this book, each new puzzle presents some new challenges and, if you think about how to refine the techniques to overcome these challenges in the most efficient way, you will find that not only are you solving every puzzle, but also your times are steadily improving. Think about the constraints on the content of each cage, and about inverting the normal logic, i.e. not only what can go in somewhere, but also what cannot. Almost every move will be difficult at this level, so do not be put off when it is hard going.

The second step is to improve your times. To do this, focus particularly on visualizing the cage integration that is used to start the puzzle, and the summation combinations that are the main tool thereafter, so that your brain can do most of this processing work subconsciously. Above all, practice makes perfect in Su Doku, and you cannot expect to achieve sub-20 minute times without having solved hundreds of puzzles.

	A	B	C	D	E	F	G	H	I
1	6		10		29	8	10	22	
2	21					(10)			
3	15		(8)	16	(8)	23 (10)			
4					3	2	14 ⁹ ₇	15	
5	27				1	3	14		
6	17	22			13	21 ⁹ ₇			
7			22			14	17		16
8		5		27				6	
9					8		6		

Fig.1

Cage Integration

You will already know that the sum of all the digits in a row, column or region is 45 so, if as in the bottom right region, all but one of the cells are in cages that add up to 39, the remaining cell G9 must be 6. This concept can be extended to solve a number of cells and get the puzzle started. It is the only way to start a Killer at this level, and it is essential, if the rest of the puzzle is to be solved, to

find every single cell and pair of cells that can be solved in this way at the start. The others in the example are solved as follows:

- In the top right region, the cages add up to 55. Two cells (F2 and F3) are sticking out, and therefore make up a pair that adds up to 10 – indicated by (10) in each.
- The cages in columns A, B, C and D add up to 188 with E9 sticking out, so E9 must be 8. It is typical at this level to have to add up three or four rows or columns.
- Row 5 has two long thin cages which add up to 41, leaving a pair (E5 and F5) that must add up to 4, i.e. 1 and 3. As the cage with sum 3 has to contain 1 and 2, E4/E5/F5 must be 2/1/3.
- The cages in rows 1 and 2 add up to 98, so the pair sticking out (C3 and E3) must add up to 8.
- Column F can be added up because F1 is solved, F2 and F3 add up to 10, F5 is solved, and $F7+F8+F9=8$ because they are the remnants of a cage of 14 where the cell that sticks out is already resolved as 6. So F4 and F6 are a pair that add up to 16, for which the only combination of digits is 7 and 9.

To get really good at this, practise visualizing the shapes made by joining the cages together to see where they form a contiguous block with just one or two cells either sticking out or indented. You also

need to practice your mental arithmetic so that you can add up the cages quickly and accurately.

Combination elimination

The main constraint in Killers is that only certain combinations of digits are possible within a cage. Easier puzzles rely on cages where only one combination is possible, e.g. 17 in two cells can only be $8+9$. At this level, however, there will be very few easy moves, and most cages will have multiple possible combinations of digits. It is necessary to eliminate the combinations that are impossible due to other constraints in order to identify the one combination that is possible. This concept is used to solve the bulk of the puzzle.

It is useful to identify where the possible combinations all contain the same digit or digits, so you know that a digit has to be somewhere in the cage. The digit can then be used for scanning and for elimination elsewhere. It is also useful to identify any digit that is not in any of the combinations and so cannot be in the cage.

The most popular multiple combinations are:

Two cell cages	Three cell cages	Four cell cages
$5 = 1+4$ or $2+3$	$8 = 1+2+5$ or $1+3+4$ (always contains 1)	$12 = 1+2+3+6$ or $1+2+4+5$ (always contains 1+2)
$6 = 1+5$ or $2+4$	$22 = 9+8+5$ or $9+7+6$ (always contains 9)	$27 = 9+8+7+3$ or $9+8+6+4$ (always contains 9+8)
$7 = 1+6$ or $2+5$ or $3+4$		$28 = 9+8+7+4$ or $9+8+6+5$ (always contains 9+8)
$10 = 1+9$ or $2+8$ or $3+7$ or $4+6$		
$11 = 2+9$ or $3+8$ or $4+7$ or $5+6$		
$12 = 3+9$ or $4+8$ or $5+7$		
$13 = 4+9$ or $5+8$ or $6+7$		
$14 = 5+9$ or $6+8$		
$15 = 6+9$ or $7+8$		

	A	B	C	D	E	F	G	H	I
1	6		10 7	1 2	29 5 9	8	10	22	
2	21			1 2	5 7 9	6 4			
3	15		(8)	16 3	(8) 5 7 9	23 6 4			
4	3 1	3 1	5 8	5 8	3 2	14 7	4	15 9 6	9 6
5	27 4 6 8 9	4 6 8 9	4 6 8 9	4 6 8	1	3	14 2 5 7	2 5 7	2 5 7
6	17	22			13 4 6	21 9	1 3 8	1 3 8	1 3 8
7			22		3 4 6	14 1 2 5	17		16
8		5		27	3 4 6	1 2 5	6		
9					8	1 2 5	6		

Fig.2

Working from the last example, the following moves achieve the position above:

- F7/F8/F9 form a cage with sum 8; 1+3+4 isn't possible because of the 3 in F5.
- F1/F2 has sum 10 and 4+6 is all that remains.
- Cage A5–D5 cannot be 9+8+7+3 because of the 3 in F5.
- Cage G5–I5 must therefore be 2+5+7, so cage

H4/I4 cannot contain 7 and must be 6+9.

- The remaining empty cells in the centre right region are G6–I6 which must be 1+3+8, so F6 is 9 and F4 is 7.
- E1–E3 must be 5+7+9 because a cage of four cells with sum 29 has only one possible combination. So E6–E8 must be 3+4+6 (but E6 cannot be 3).
- D3 has to be 1 or 2 or 3. It cannot be 1 because this would make C4/D4 a pair with sum 15 and neither of the combinations for this is possible because of the other digits already in row 4. The same applies if D3 is 2. Therefore, D3 has to be 3, and the only possible combination for C4/D4 is 5+8.
- D1/D2 are now 1+2, making C1=7.

Getting good at this is rather like learning the times table at school, because you need to learn the combinations off by heart; then you can just look at a cage and the possible combinations will pop into your head, and you can eliminate the ones that are excluded by the presence of other surrounding digits.

	A	B	C	D	E	F	G	H	I
1	⁶ ₂ 4	⁴ ₂	¹⁰ 7	1	²⁹ 9	8	¹⁰ 3	²² ₅ 6	₅ 6
2	²¹ ₈ 3	₈ 3	₈ 3	2	5	6	1	₄ 7	₄ 7
3	¹⁵ ₅ 6	₅ 6	1	¹⁶ 3	7	²³ 4	₂ 9	₂ 8 9	₂ 8 9
4	₁ 3	₁ 3	₅ 8	₅ 8	³ 2	¹⁴ 7	4	¹⁵ ₆ 9	₆ 9
5	²⁷ ₈ 4 6	₈ 4 6	₈ 4 6	₄ 6 8	1	3	¹⁴ ₅ 2 7	₅ 2 7	₅ 2 7
6	¹⁷	²²			¹³ ₄ 6	²¹ 9	8	₁ 3	₁ 3
7			²² ₆ 9	7	₄ 3 6	¹⁴ ₂ 1 5	¹⁷		¹⁶
8		₅	₆ 9	²⁷ 6	₄ 3 6	₂ 1 5	₆		
9			4	9	8	₂ 1 5	6		

Fig.3

Further combination elimination

To progress from where the last example finished off to the position above:

- Go back to the fact that $C3 + E3 = 8$, as marked by the (8) symbols; given the constraints already on E3 and the 3 in D3, the only possible combination is $C3 = 1$ and $E3 = 7$.

- As $C3=1$, the cage $A1/B1$ can only be $2+4$, which also resolves $D1$ and $D2$.
- $A3+B3=11$, for which the only combination now possible is $5+6$; which makes $F3=4$ and $F2=6$, so $G3-I3$ is $2+8+9$.
- The cage $C7/C8/D7$ can only be $9+8+5$ or $9+7+6$, but if it were $9+8+5$ then $D7$ would have to be 9 , because 5 and 8 are already in the bottom centre region; so $C7/C8$ would have to be $5+8$, which is not possible because $C4$ is also 5 or 8 . So $D7=7$ (because column C already has a 7) and $C7/C8$ is $6+9$.
- The adjoining cage, $C9/D8/D9/E9$, contains $8+9$ and either $3+7$ or $4+6$, but it cannot contain 7 , so it must be $8+9+4+6$.

The last two cages were something that I had been looking forward to from the start, because they both contain the same sorts of digits and are closely intertwined, so it had to be that a single key ($C4$ as it turned out) would unlock the whole lot.

	A	B	C	D	E	F	G	H	I
1	⁶ 2	4	¹⁰ 7	1	²⁹ 9	8	¹⁰ 3	²² ₅ ⁶	₅ ⁶
2	²¹ 8	9	3	2	5	6	1	₄ ⁷	₄ ⁷
3	¹⁵ 6	5	1	¹⁶ 3	7	²³ 4	² ₉	₈ ² ₉	₈ ² ₉
4	3	1	5	8	³ 2	¹⁴ 7	4	¹⁵ ₆ ⁹	₆ ⁹
5	²⁷ 9	6	8	4	1	3	¹⁴ ₅ ² ₇	₅ ² ₇	₅ ² ₇
6	¹⁷ 4	²² 7	2	5	¹³ 6	²¹ 9	8	₁ ³	₁ ³
7		8	²² 6	7	₃ ⁴	¹⁴ ₂ ¹ ₅	¹⁷ ₉	₉	¹⁶
8		₂ ⁵ ₃	9	²⁷ 6	₃ ⁴	₂ ¹ ₅	₆		8
9		₂ ³	4	9	8	₂ ¹ ₅	6		

Fig.4

Inverse logic

The example shows a very interesting situation in the bottom right region – the cage with sum 17 is missing two digits that add up to 8, and so is the cage with sum 16. The possible combinations are 1+7, 2+6 and 3+5, but 2+6 is not possible because G9 is already 6, so the 2 must be in the cage H8/H9. This provides the final breakthrough to finish the puzzle.

If you get stuck at any point, and find yourself having to contemplate complex logic to progress, it will almost certainly be because you did not find all the cage integration opportunities at the start. So the best way to get going again is to look for more cage integration opportunities.

Finally, keep looking out for opportunities to use classic Su Doku moves wherever possible, because they will be relatively easy moves. Good luck, and have fun.

Mike Colloby
Captain, UK National Su Doku Team 2010

Puzzles

Deadly

W

u

✓

1

4 42

2 3 15 23 894
2 3 6 5 7 9 8 7 4

5 3	2	11 6 5	16 7	9	15 8	12 1	4	
21 4	9	8	18 6	1	3	7	5	2
6 5	1	18 7	8	23 8	4	3	18 6 9	
15 1	5	4	2	8	7	6	9	3
6 6	8	5 3	7 4	9	5	18 1	2	7
22 9	7	2	3	18 6	8 1	5	8 4	8
15 7	6	14 5	9	4	8	2	3	1
8	12 3	9	12 1	5	2	4	18 7	6
7 2	4	1	10 7	3	23 6	9	8	5

1 2 4 3 7 6 8 9 5

⌚ 1 HOUR

TIME TAKEN.....

~

Deadly

12 ~~34~~
45

1236
1245

28	12			10			22	
			14	21		19		
12		14			9			
						21	23	
17		22						
		21		10	8		7	
21					3	8	22	5
15	12	13	15	8	13	19	4	
	7	14	28			19	4	

⌚ 27
1 HOUR

TIME TAKEN.....

9568 8 13
2367 3 23
6859
47

M

9876
 591713 9579
 68 7864
 (59) 91234
 (18) 9 (124)

14	29	30			17	7		
17				9	3	3	25	
16		11	9		21	3		
						20		17
	21	28			14			
11						21		
		22	16		22			
	9					8	24	
14								9

432 1 789 765 876
 1 HOUR 754321 1/2 3456 789
 TIME TAKEN..... 87654 76543
 654 537 876 5678
 876 9- 3967 3578 956
 987- 3589 978
 65 918 23456
 Deadly 978

11				10		22		17
20			21			12		
21				16				
12	14		10				10	20
	12			13		20		
		13		7				
12	22	10	22		13			
					20			9
		7			9			

⌚ 1 HOUR

TIME TAKEN.....

68 15

14 8	6	6 1	5	21 3	13 4	9	14 2	7
7 5	22 9	7	6	1	2	10 4	19 3	8
5	7 3	4	3 2	21 7		6	1	23 5
10 4	8 7	18 5	1	6	5 2	3	8	9
6	1	9	4	8	9 3	5	7	2
23 3	13 8	8 2	19 9	12 5	7	1	18 6	4
9	5	6	7	2	1	8	4 4	3
7	4	11 3	17 8	9	13 6	2	5	1
3 1	2	8	7 3	4	27 5	7	9	6

1 2
⌚ 1 HOUR

3 4

TIME TAKEN.....

765

957

8765

9756

98

97568

457

Deadly

29	19			10	14	17		8
		21						
18			15				21	19
			21		20			
	17		9					
19						9		19
	12	10	18			14		
				20	7			
					19			

⌚ 1 HOUR 5 MINUTES

TIME TAKEN.....

15	21	9			23	23		10
		12						
			12		6	20		
23		9	17				11	
	21			17	10		14	
			11		21	10		14
8	26	8						
		3					16	
			13		10			

⌚ 1 HOUR 5 MINUTES

TIME TAKEN.....

3

12 3 6

6789

9	15	5		25	5	11		29
						10		
16	21		12	12				
				8	13		26	
	19							16
	9	12			29		17	
20		8						
			11	12			19	
16								

⌚ 1 HOUR 5 MINUTES

TIME TAKEN.....

11	24	16			11			15
				10			7	
22		4		14	17		3	
		14				8	22	13
10				17				7
	17				12			
17	8		20			9	4	5
	12		20			25		10
13			5					3
								7

9
6
5
8
1
2
4
3
7

⌚ 1 HOUR 5 MINUTES

TIME TAKEN.....

3746

1897
2689

12
17

Deadly

11	25		20		10			
				12	17		11	
8	12		8		27		18	
				22				
13		20	3		19			11
	28					14		
			18			16	8	
	13				15			14
		12						

⌚ 1 HOUR 5 MINUTES

TIME TAKEN.....

19				17		10	3		
20		18		11			25		
				4	5	9			
16		23					11		
22				9		17	23		
	10	21						12	
		22	8				12		
8			16		23				
4			7			9	4		

2 5

1 3

⌚ 1 HOUR 5 MINUTES

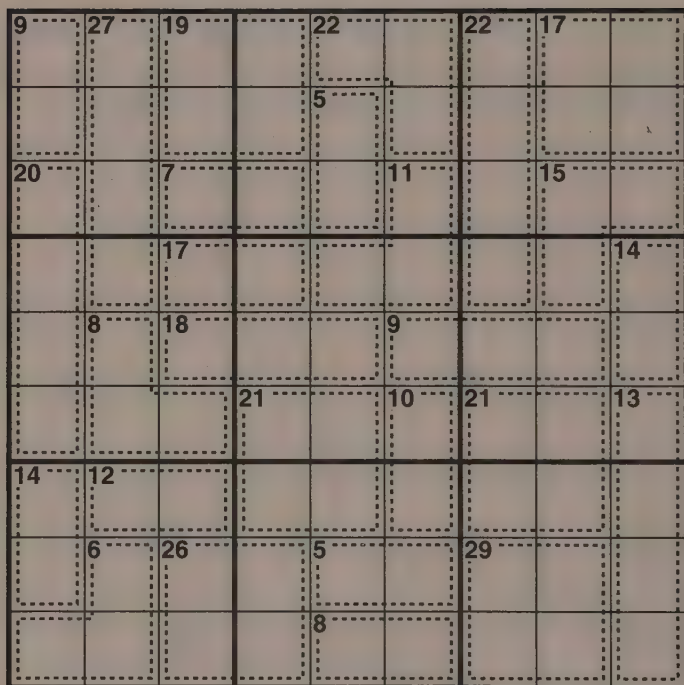
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134

79

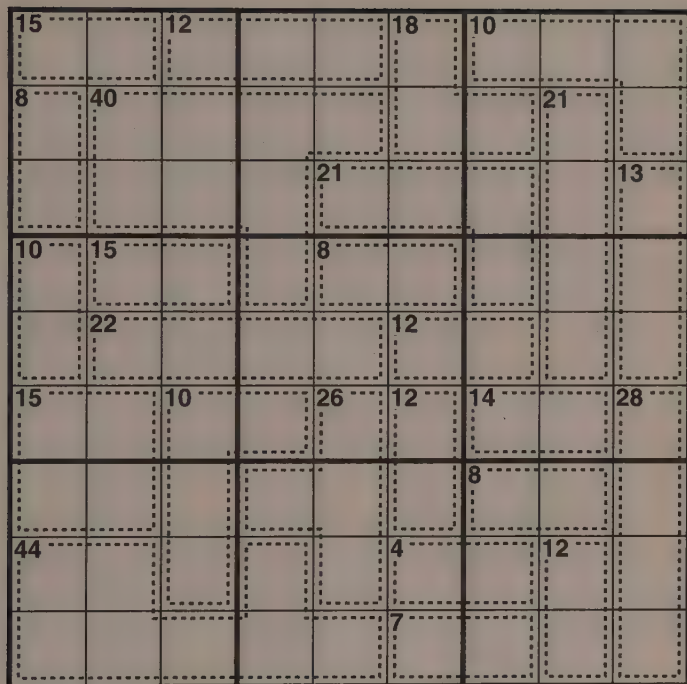
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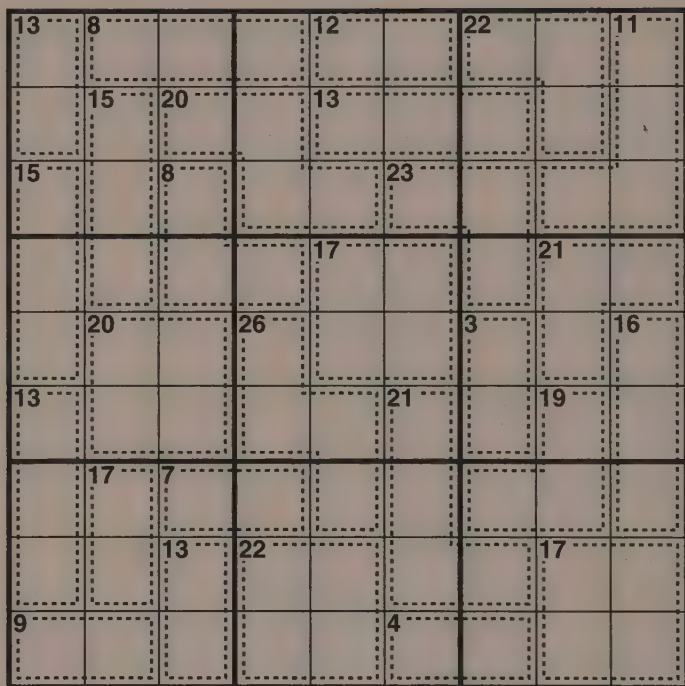
⌚ 1 HOUR 5 MINUTES

TIME TAKEN.....



⌚ 1 HOUR 5 MINUTES

TIME TAKEN.....



⌚ 1 HOUR 5 MINUTES

TIME TAKEN.....

9	30			17				
		9		14		23	10	
22		15			21			18
						20		
10			12					
15		18		21		15	23	
	7				18			
		10					14	
14			20					

⌚ 1 HOUR 5 MINUTES

TIME TAKEN.....

14	17	13			9			18
		7	11	24				
					19	22		
18	18		3			9		
	10				22		22	
		28		8		12		20
21			13		9		6	
9				23				

⌚ 1 HOUR 10 MINUTES

TIME TAKEN.....

17	19			7	23		10	
			25				28	
8	10				19	5		
			24				4	
	15				12			18
11		10		22		7		
20		7					14	
			17	9	3	21		7
13								

⌚ 1 HOUR 10 MINUTES

TIME TAKEN.....

24	17		21		14		33	
						17		
			17	20				
3					21	20		
10		20						
8	15				10	19	11	
		30	17				21	
			16					
				9			12	

⌚ 1 HOUR 10 MINUTES

TIME TAKEN.....

20		7		13			19	
13			20		9			
	9			9	23	9	24	
		16						
24				4			13	
20					3	15		15
20		14						
	11	6	9	16		20		
				13			11	

⌚ 1 HOUR 10 MINUTES

TIME TAKEN.....

8		12			19		25	
	22		10	24				
24						9		
	9		13		9		22	4
		21		20	9			
9		8				17		8
7					22			
11		15		10			7	15
	16							

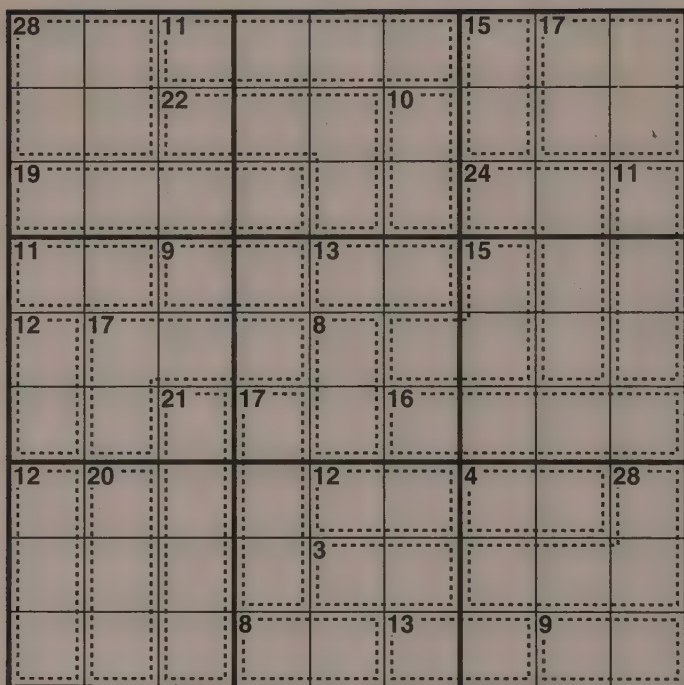
⌚ 1 HOUR 10 MINUTES

TIME TAKEN.....

18	23			7	7			10
		20			24			
3		12		11	20	13		
						7		30
13		21	8			12		
15	7			12		3		
			20		17			
10					8		9	
16			11		18			

⌚ 1 HOUR 10 MINUTES

TIME TAKEN.....



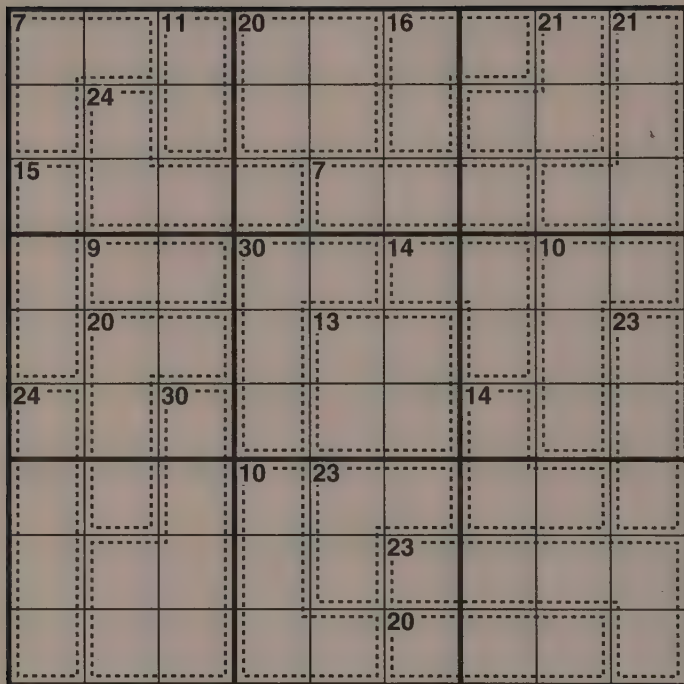
⌚ 1 HOUR 10 MINUTES

TIME TAKEN.....

29			17		22	22		
17	20				13	16		7
			8	29		22		
21		19				17		
			15					
						19		16
24		9	30		8			
							5	

⌚ 1 HOUR 10 MINUTES

TIME TAKEN.....



🕒 1 HOUR 10 MINUTES

TIME TAKEN.....

15	16		12	17			10	
		14			18			30
6			12					
21						23	12	
		6	12				8	
19	10		28		20			
					7			14
	11	20			14	19		
			11					

⌚ 1 HOUR 10 MINUTES

TIME TAKEN.....

22	13		21		20	16		12
	9							
		4	20			9		13
24					12		10	
	12			8	13			
	12		18			20		
12		21				21		
			24	11				
8					20			

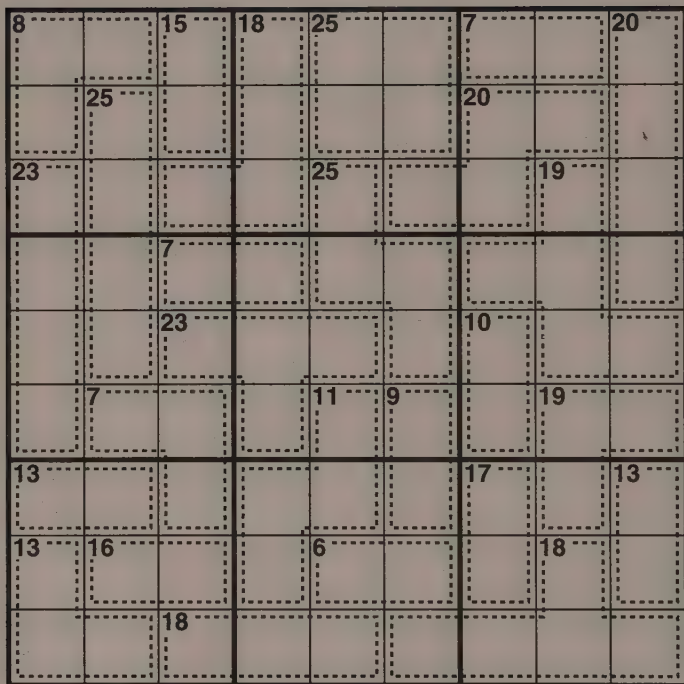
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TIME TAKEN.....

23			14		9		20	
	17	17		24				11
19			28			9		
					16	20		18
28	7	8	20	22				
					19			20
	8			11			17	

⌚ 1 HOUR 10 MINUTES

TIME TAKEN.....



⌚ 1 HOUR 10 MINUTES

TIME TAKEN.....

11	14			7		24	
	22	15		22			10
5					17	11	
	10		22	9			
28	19						13
			17	21	10	20	20
	9	6					
					10	11	
16						6	

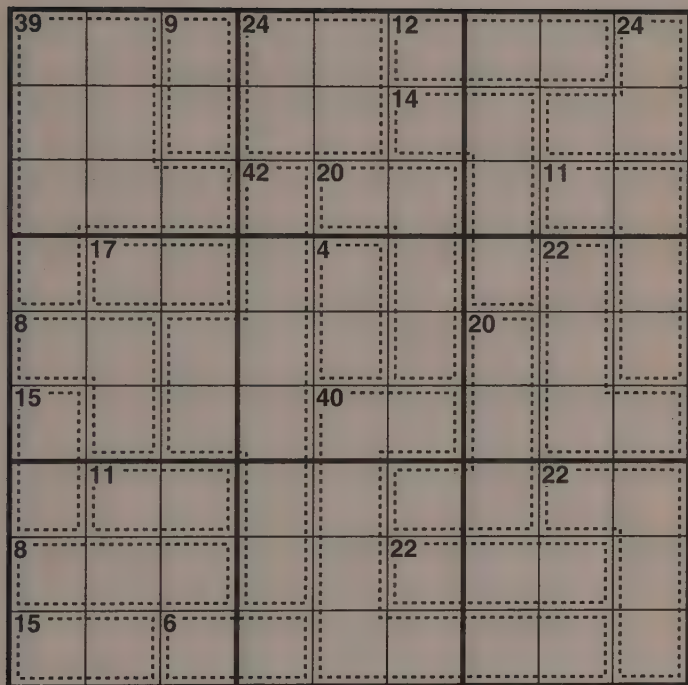
⌚ 1 HOUR 10 MINUTES

TIME TAKEN.....

7		27	21	24				8
15						13		
			7	12			23	
23					19			
	17						10	
21		6	35	10	19		23	
26						23	10	
	6							

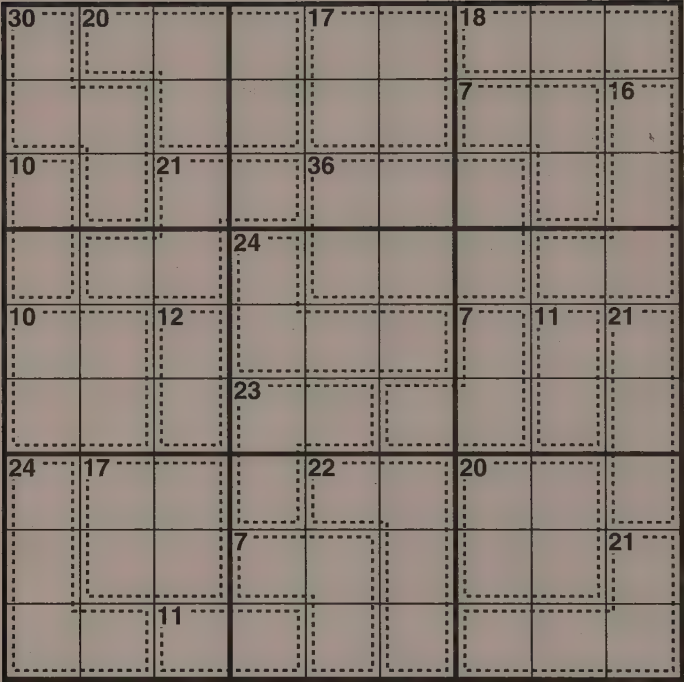
⌚ 1 HOUR 15 MINUTES

TIME TAKEN.....



⌚ 1 HOUR 15 MINUTES

TIME TAKEN.....



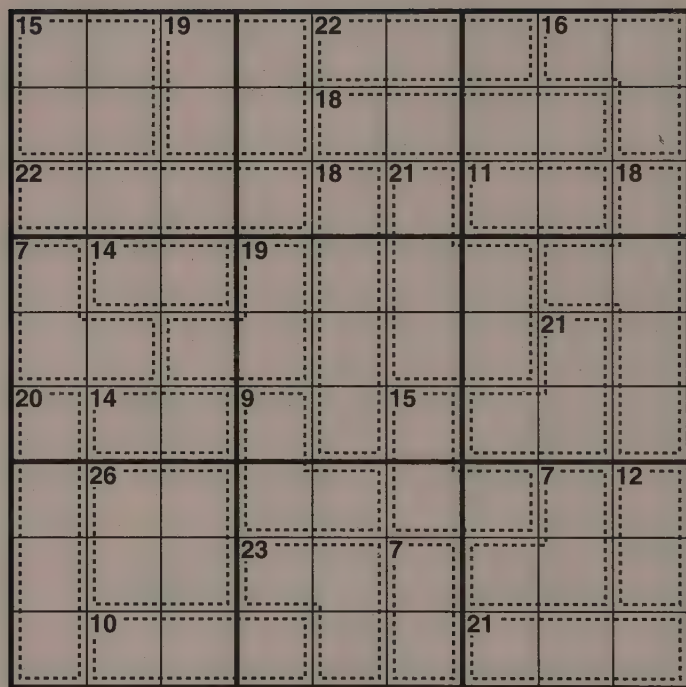
⌚ 1 HOUR 15 MINUTES

TIME TAKEN.....

27		15			8	25		
8			8				16	
	8	14		17		17		
			21				22	
30					10			
			12		14		20	11
7		15				12		
18			21	12				13
					4			

⌚ 1 HOUR 15 MINUTES

TIME TAKEN.....



⌚ 1 HOUR 15 MINUTES

TIME TAKEN.....

10	8	22				7		16
		26		17				
19					20	25		
	28						19	
			17		27			
10							9	
	9		28		18			11
7					11			
24					17			

⌚ 1 HOUR 15 MINUTES

TIME TAKEN.....

17		13			30	18	20	
		22	8					
								10
22	10		25					
			20		9	20		22
7	10	20		13				
						32		
11	16							
		18					12	

⌚ 1 HOUR 20 MINUTES

TIME TAKEN.....

6	20			17		8		32
		12						
19	20				19		8	
	22		12				3	21
				15	22			
12		16	8			8		
19					24		16	4
	7		11					
					8		16	

⌚ 1 HOUR 20 MINUTES

TIME TAKEN.....

9	21		15	11	16			10
						8		
13		22			27		9	8
23		12						
		10		13		25		
7		11			8			17
12					13			
22		20		4		19		
					12		8	

⌚ 1 HOUR 20 MINUTES

TIME TAKEN.....

22		14		21			9
	17	11		14			
			18		16	30	
	7	22		20		10	
32		11	21	18			
				18		12	
		21			20		
9				3		9	

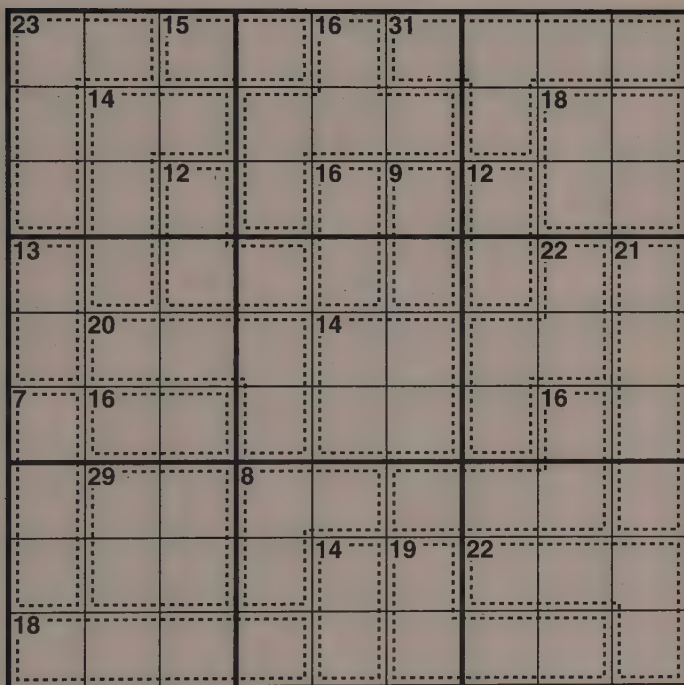
⌚ 1 HOUR 20 MINUTES

TIME TAKEN.....

27		11			14	10		
			11	21		29		20
20							17	
22								
14		20		18	21	11		
	19						3	
						17		10
	30			12				
8					20			

⌚ 1 HOUR 20 MINUTES

TIME TAKEN.....



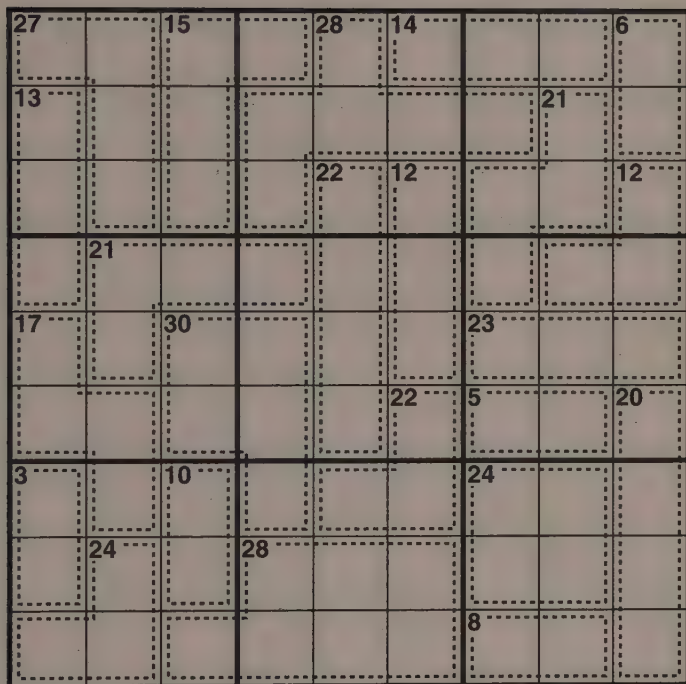
⌚ 1 HOUR 20 MINUTES

TIME TAKEN.....

12	10		25	11			15	
	22			11				23
24				13				
					22		16	
19		10		25				
7		19				9	8	
			13				21	
22					16			4
		16			12			

⌚ 1 HOUR 20 MINUTES

TIME TAKEN.....



⌚ 1 HOUR 20 MINUTES

TIME TAKEN.....

9		22				18		16
20		8		11			22	
		15			10	4		
9		14	9	21				
21	8						21	10
			14					
	19				7	10		14
	19		17			12		
			13		12			

⌚ 1 HOUR 20 MINUTES

TIME TAKEN.....

24			12			21	9	
	16	20	11					16
16					23	10		
		10						
		24	11	10	20		13	
							11	
17	10		17		22			8
	8					9		
8		17					12	

⌚ 1 HOUR 20 MINUTES

TIME TAKEN.....

21	13	21			15			
			10	8	9		16	11
	7					18		
7	4	23	12				7	
				13		8		20
13	23		3		8		20	
			18			16		
4	9	17						
			16		6		9	

⌚ 1 HOUR 20 MINUTES

TIME TAKEN.....

20	6		16	17		17		19
						9		
11	19			20	4		10	
			19		9			
17						15		22
22	8			20				
	14	9					15	
		15	13		24			15

⌚ 1 HOUR 20 MINUTES

TIME TAKEN.....

7		25				23	9
18			31				
15		32				15	
					22		4
22	15		15				
	25				13		11
			17	10			
	16			29		7	15
		9					

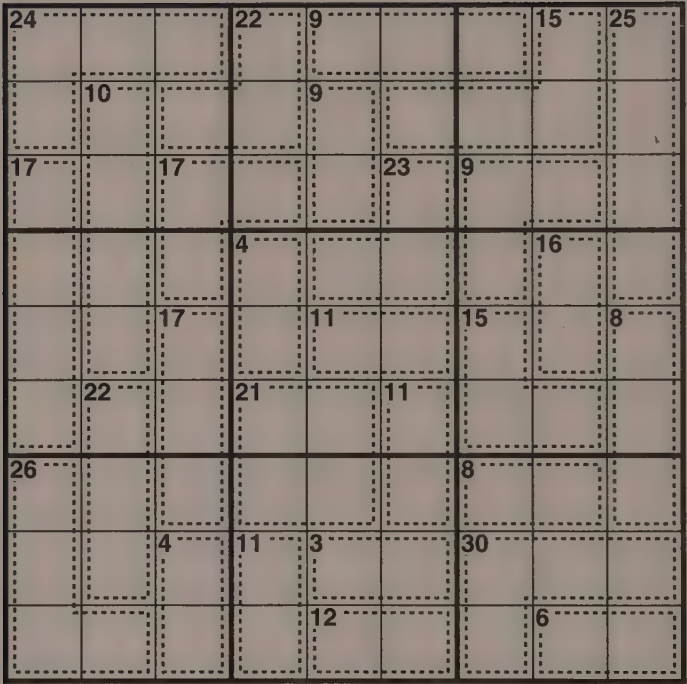
⌚ 1 HOUR 20 MINUTES

TIME TAKEN.....

18		11		14			25	
			24	13				
21		13				17		
			25	12			17	
7				11				
	8	14		22	17		7	13
			12					
22	12			11	8	14		
							17	

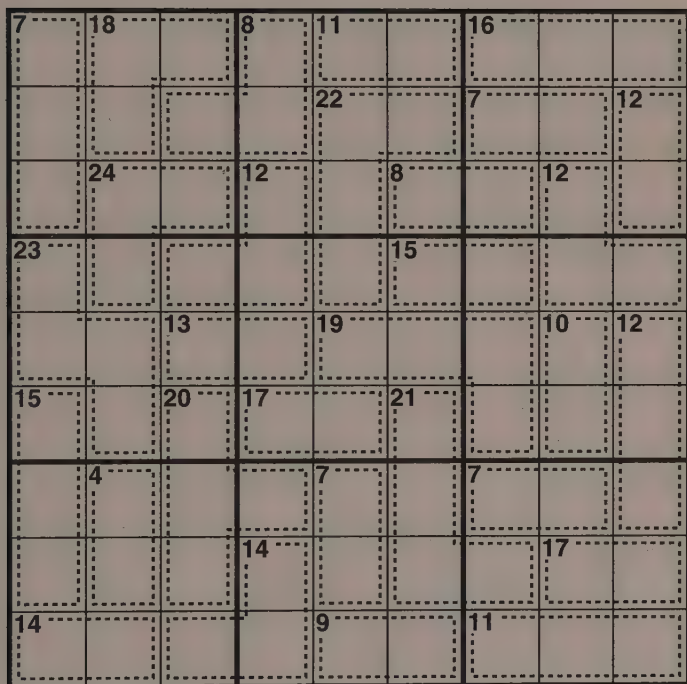
⌚ 1 HOUR 20 MINUTES

TIME TAKEN.....



⌚ 1 HOUR 20 MINUTES

TIME TAKEN.....



⌚ 1 HOUR 20 MINUTES

TIME TAKEN.....

11		22		19		17		
20			9				8	
				9		19		8
9	16	20			14			
		20				11	20	
4	25		18	9				21
					22			
9			9					12
	8			13		3		

⌚ 1 HOUR 25 MINUTES

TIME TAKEN.....

16		24		21			
			14	7		25	
8		9		4	22	18	
19	7						9
		14		16		17	
11	17	16		13		9	
			5			19	
5	12	5		23			
				13		7	

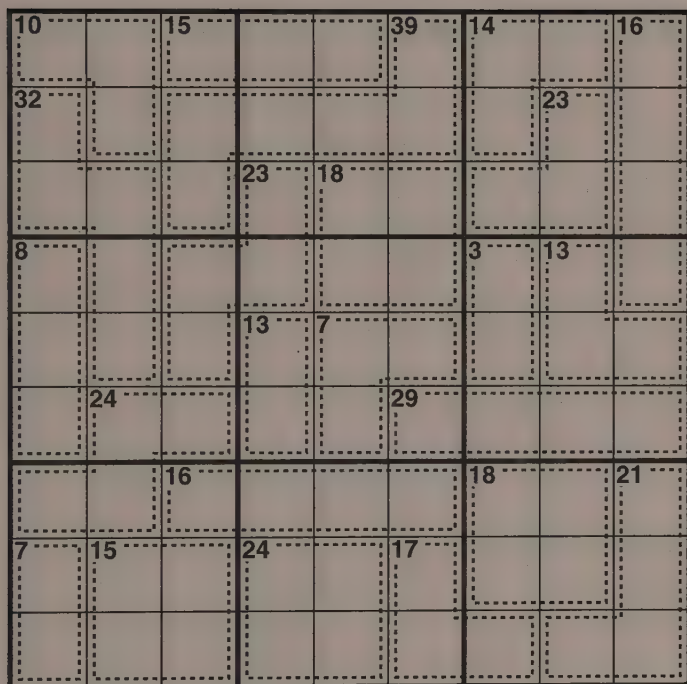
⌚ 1 HOUR 25 MINUTES

TIME TAKEN.....

20		18	25			25		
7			14				18	
				12	11	5		
16	10	3					20	7
			13	21		11		
14		19		21				13
			12			21		
11	9							
		12			3		14	

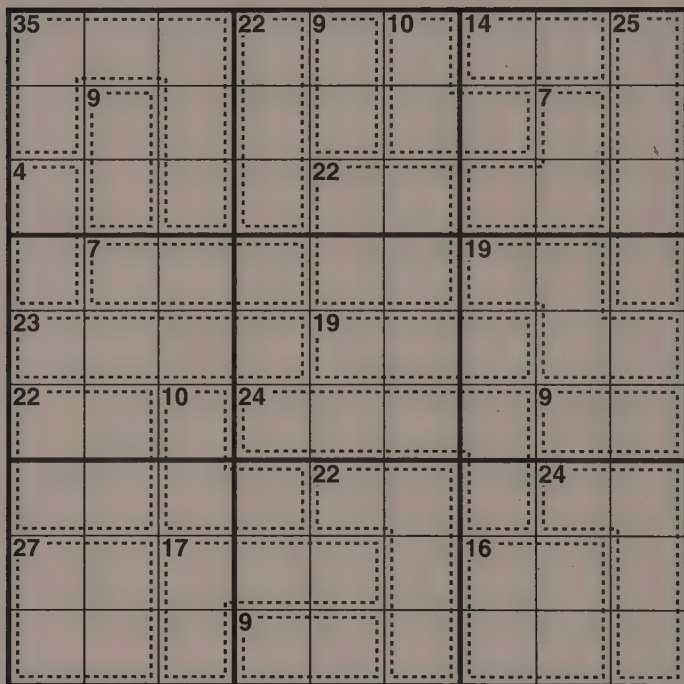
⌚ 1 HOUR 25 MINUTES

TIME TAKEN.....



⌚ 1 HOUR 25 MINUTES

TIME TAKEN.....



⌚ 1 HOUR 25 MINUTES

TIME TAKEN.....

20		13		8		12		12
		11		3		11		
12		6	15	21		17		15
15					12			
	18					20	20	
18			13					
		6		22				12
21	11				28		13	

⌚ 1 HOUR 25 MINUTES

TIME TAKEN.....

17	20	22		28			21	
		2		36		10		
		6					34	
17						11		3
21								
3	1	14				26		
2	19		8	16		6	8	
19						7	4	24
4	3	1	26	9				

⌚ 1 HOUR 25 MINUTES

467

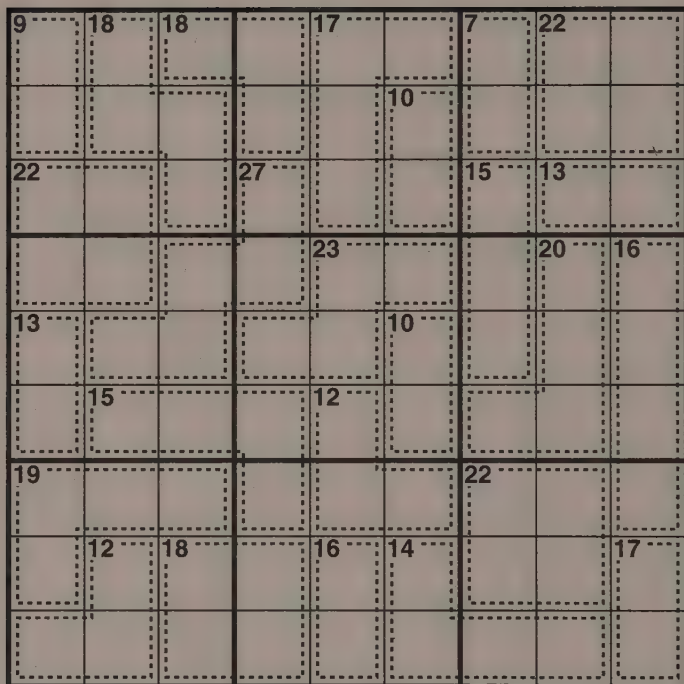
258

TIME TAKEN.....

9		19				19		
21				21	13			
15	16				22		18	
		22					16	
	9							29
14		21		13			8	
5			7		26			
7	16			7		14	6	
		12						

⌚ 1 HOUR 25 MINUTES

TIME TAKEN.....



⌚ 1 HOUR 25 MINUTES

TIME TAKEN.....

9	21			11			21	
		11		24	18			
18		20			12		6	9
20				21			19	
20	11		14		23			
	6					8		8
	11	21	8			12		
				7			16	

⌚ 1 HOUR 25 MINUTES

TIME TAKEN.....

11	12			20	11	10	20	
		30						16
25			5		21			
27				8	19		21	
	12	17				20		7
				11				
22						9		18
7		12		14				

⌚ 1 HOUR 25 MINUTES

TIME TAKEN.....

11	22			8	12		11	
	29					11	16	
			11	19	21			
23	15						12	
		26			5		10	11
		3			23			
	5			22			22	
34								
			12			11		

⌚ 1 HOUR 25 MINUTES

TIME TAKEN.....

26	5		21		8		11	
	20				18			22
		11		13		21	14	
22			10		8			
								9
6		26		26			13	
17	6			15				
		17				24		6
				10				

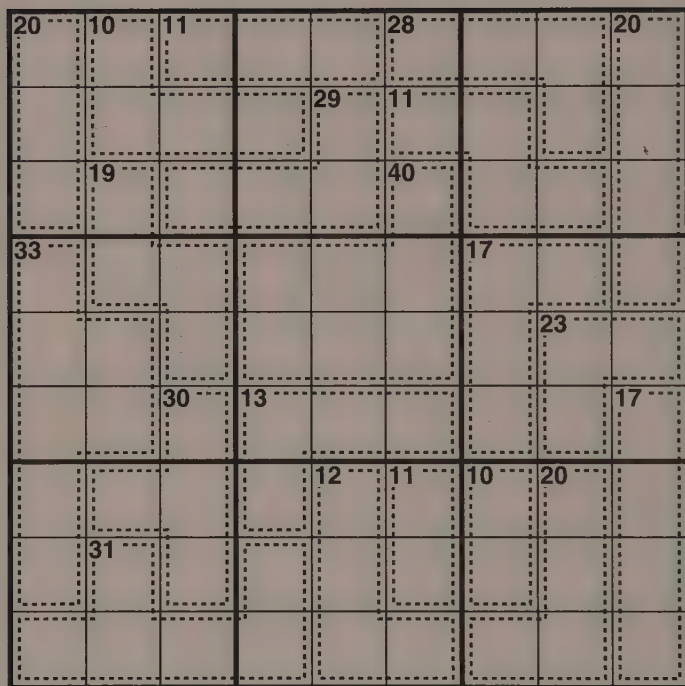
⌚ 1 HOUR 30 MINUTES

TIME TAKEN.....

22		11		7		17		
	6	17		23	19			
17					18		11	
			11	13			6	
11					8		22	
8	13		19			10		
	17		10					12
	12		27	20				
8						10		

⌚ 1 HOUR 30 MINUTES

TIME TAKEN.....



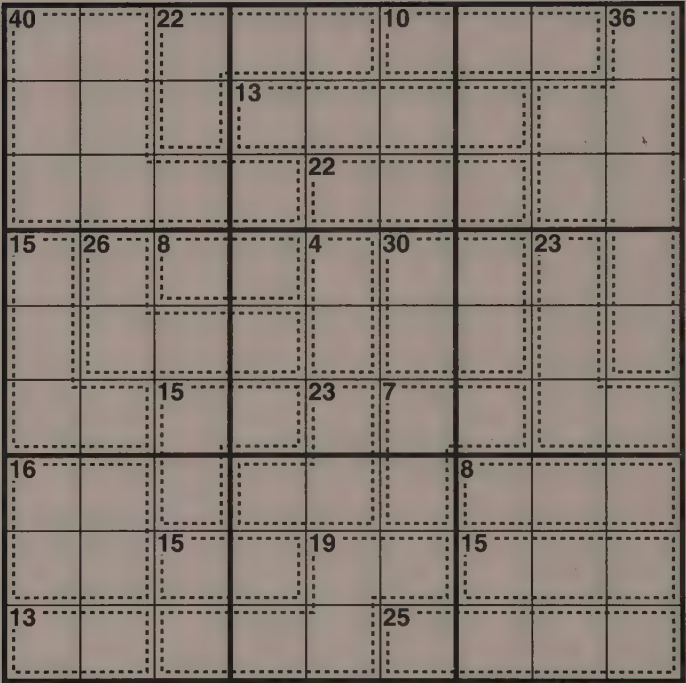
⌚ 1 HOUR 30 MINUTES

TIME TAKEN.....

17	28			16	13			
		15	10		16			
				10		21		13
9		20			17			
			15				8	
18	8			14		19		12
	20	22						
				13			16	11
7		5			12			

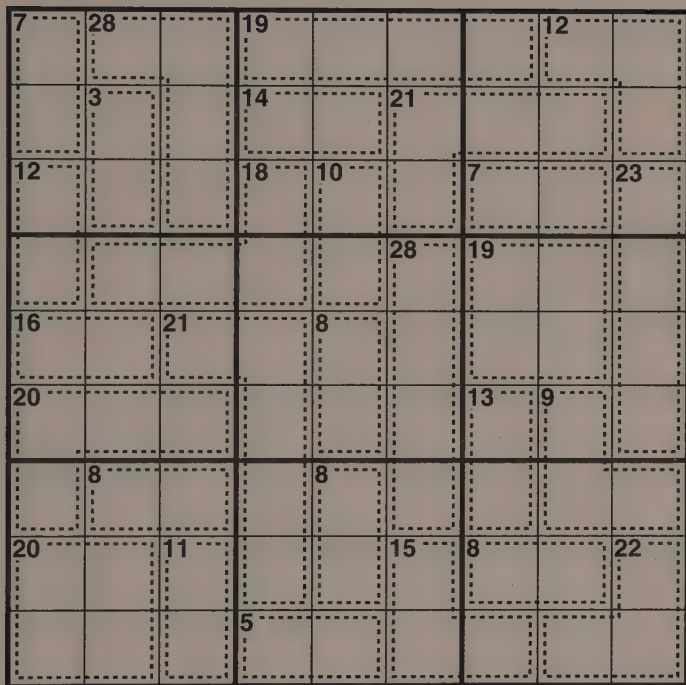
⌚ 1 HOUR 30 MINUTES

TIME TAKEN.....



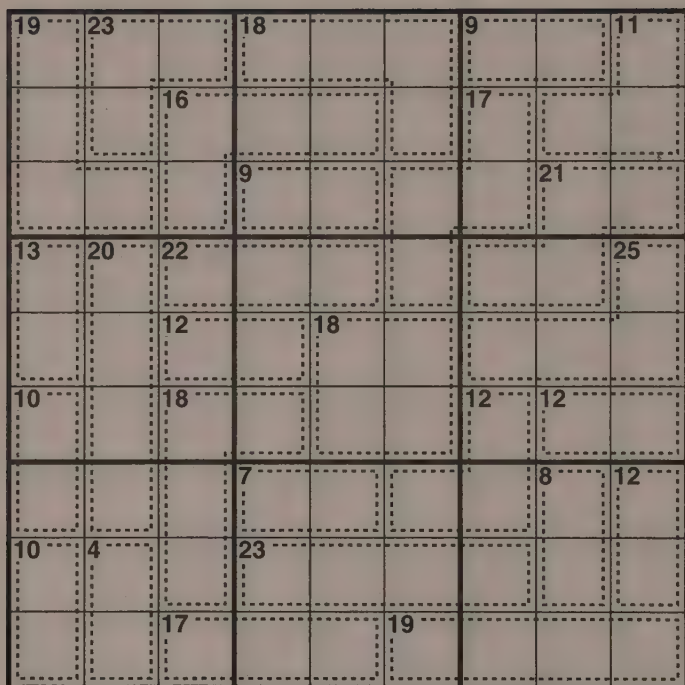
⌚ 1 HOUR 30 MINUTES

TIME TAKEN.....



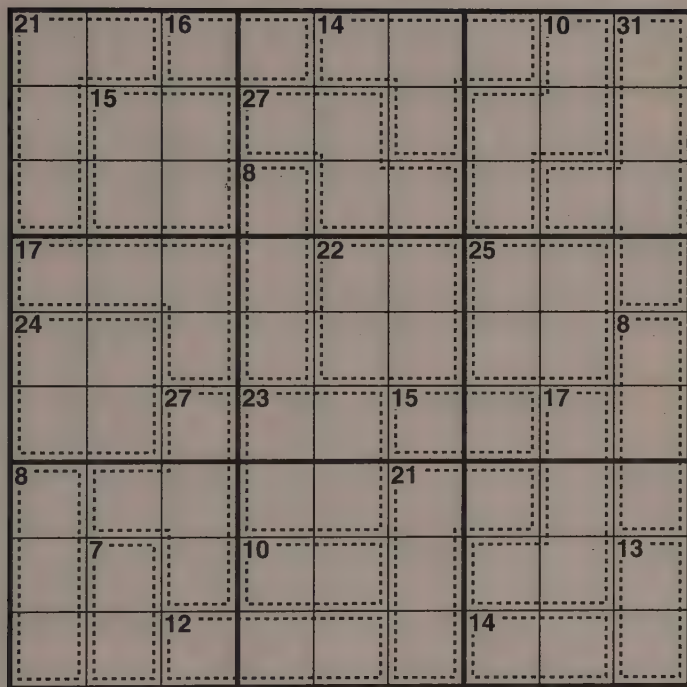
⌚ 1 HOUR 30 MINUTES

TIME TAKEN.....



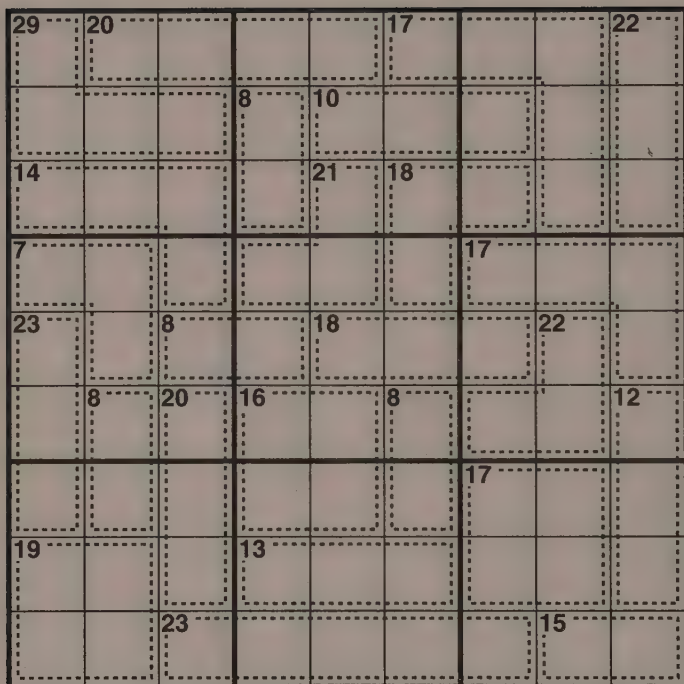
⌚ 1 HOUR 30 MINUTES

TIME TAKEN.....



⌚ 1 HOUR 30 MINUTES

TIME TAKEN.....



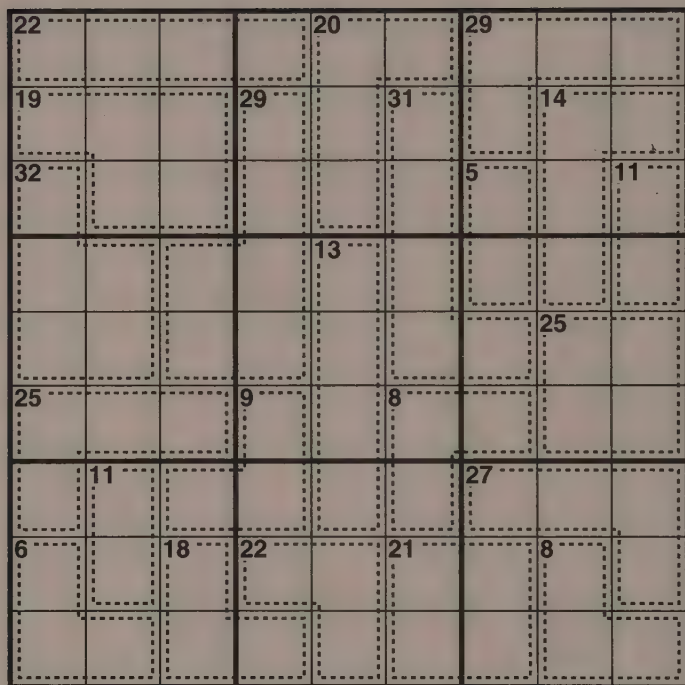
⌚ 1 HOUR 30 MINUTES

TIME TAKEN.....

14			11	20	19		24	
	25							
		20		22		29		
27								19
	23				8			
					17		22	
4	21							8
			12		22			
23						15		

⌚ 1 HOUR 30 MINUTES

TIME TAKEN.....



⌚ 1 HOUR 30 MINUTES

TIME TAKEN.....

11		5		13	24			16
24		21			8	15		
			23	19				
	20						22	
4						7		
	18	21		9	14		14	
			4					18
17		12			16			
				17		13		

⌚ 1 HOUR 30 MINUTES

TIME TAKEN.....

30	13	21		13	4		13	
		15			11			25
	19				13		8	
					20			
12	13		11			11		
		8		10	20		19	
							18	
	26	12		26				
					14			

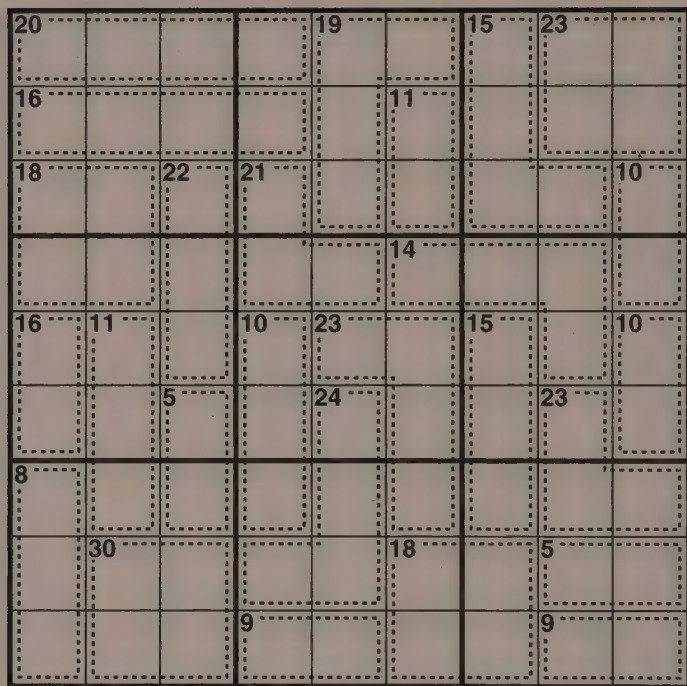
⌚ 1 HOUR 30 MINUTES

TIME TAKEN.....

5		19		19	12	24		
22							5	
12		12			23	16	11	22
	21							
12		3		11				
		15	7		30		7	
			4	11		10		6
26							16	
	9			15				

⌚ 1 HOUR 30 MINUTES

TIME TAKEN.....



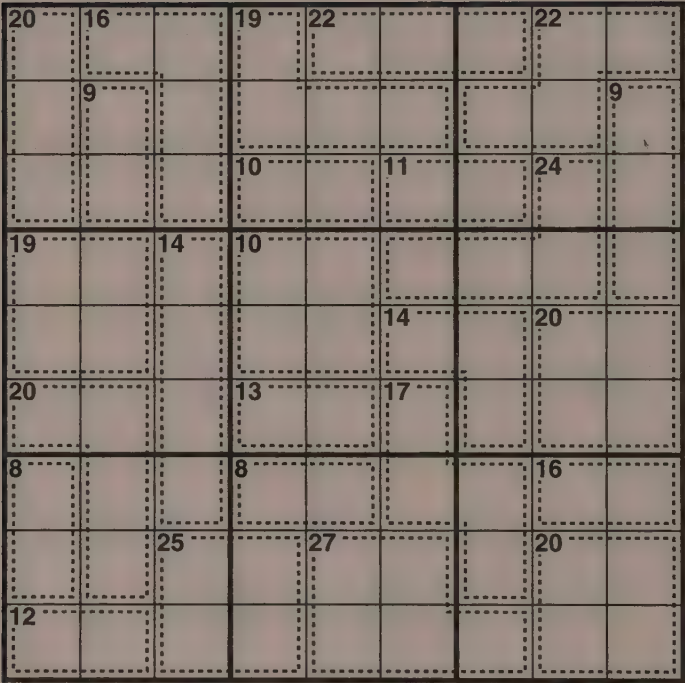
⌚ 1 HOUR 30 MINUTES

TIME TAKEN.....

25				40		20		7
19								
	6					19		27
23	8		10					
		15			21			
7		7				22	13	19
12		18						
25	16		19				7	

⌚ 1 HOUR 30 MINUTES

TIME TAKEN.....



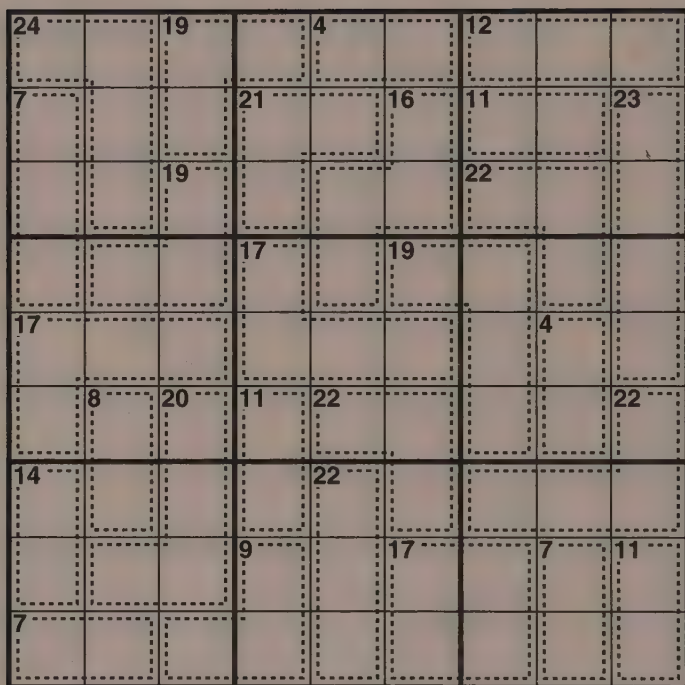
⌚ 1 HOUR 30 MINUTES

TIME TAKEN.....

10				30				21
30		25	15					
	6			10	12		29	
			23					
10		20						
			14		21		16	
8		16						
	22	13		27	21			6

⌚ 1 HOUR 30 MINUTES

TIME TAKEN.....



⌚ 1 HOUR 30 MINUTES

TIME TAKEN.....

18				17	18		18	
7	30		18					
		9			22			13
18					10			
				9	10		30	
6		25						
19					21			
22			11	22		21		11

⌚ 1 HOUR 30 MINUTES

TIME TAKEN.....

24		12		13		23		
14		23		12			16	
				23		12		29
12			16					
		18			11			
21			11	7		19		
	4				22			
			12			14	15	7
15								

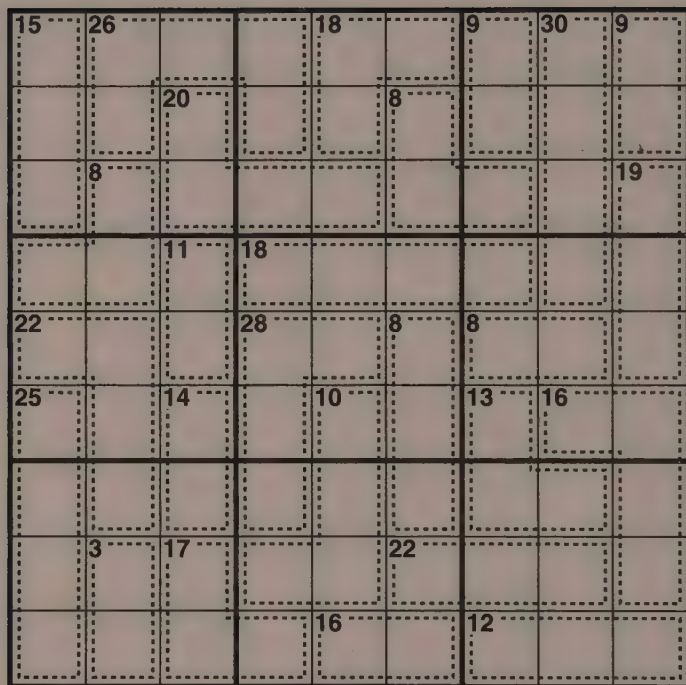
⌚ 1 HOUR 35 MINUTES

TIME TAKEN.....

21		17		19		11	23	4
	12							
12			10	28				12
	11							
4	23		20		14		13	
	19				7		13	
		22		11			11	
8	10				15		5	
		10				20		

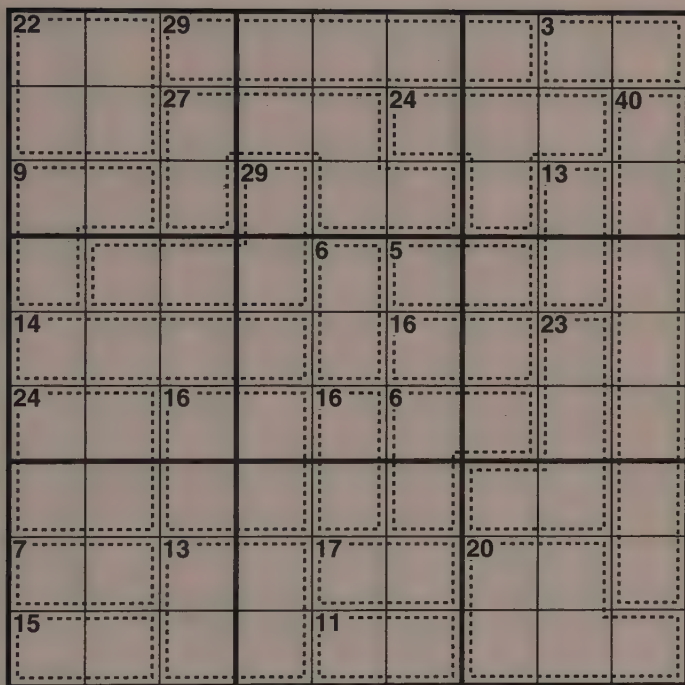
⌚ 1 HOUR 35 MINUTES

TIME TAKEN.....



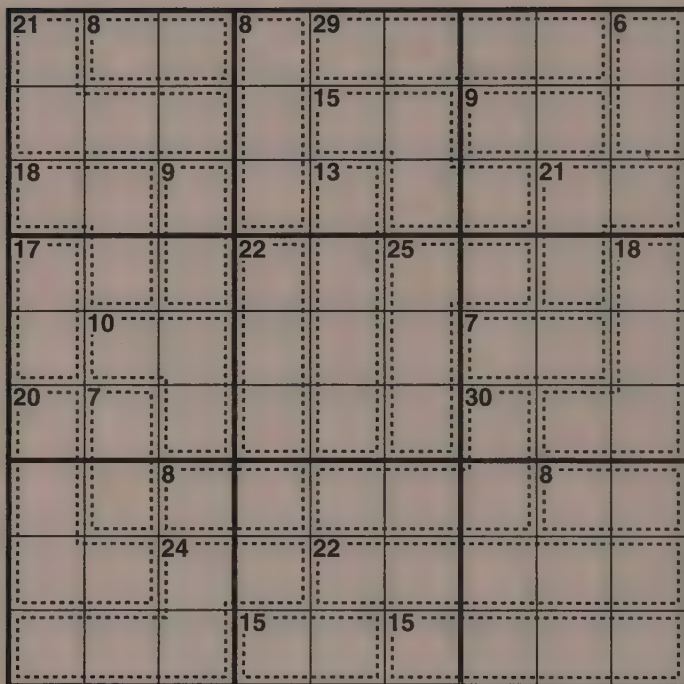
⌚ 1 HOUR 35 MINUTES

TIME TAKEN.....



⌚ 1 HOUR 35 MINUTES

TIME TAKEN.....



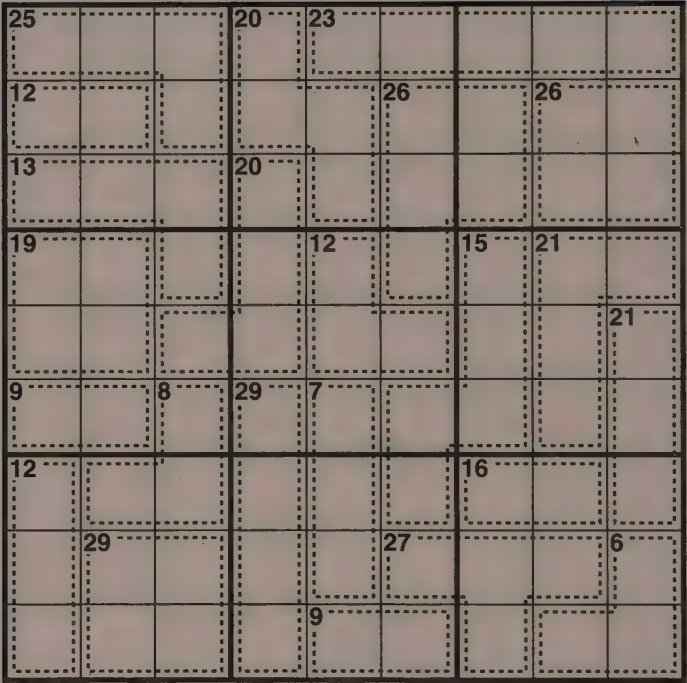
⌚ 1 HOUR 35 MINUTES

TIME TAKEN.....

18	18			27	17		19
			6				
19		23	21		11		17
21							
			18		16	13	3
	14			7			
	12			20	8		12
12		13				21	
	5			14			

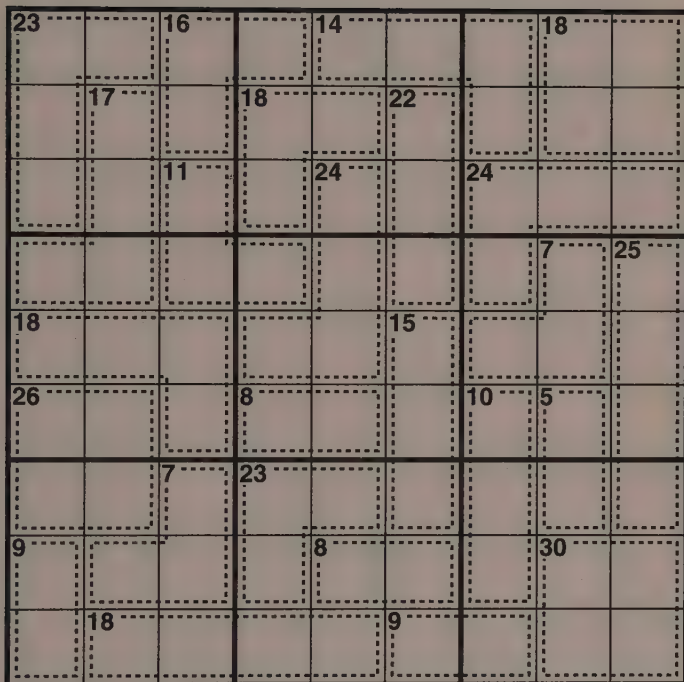
⌚ 1 HOUR 35 MINUTES

TIME TAKEN.....



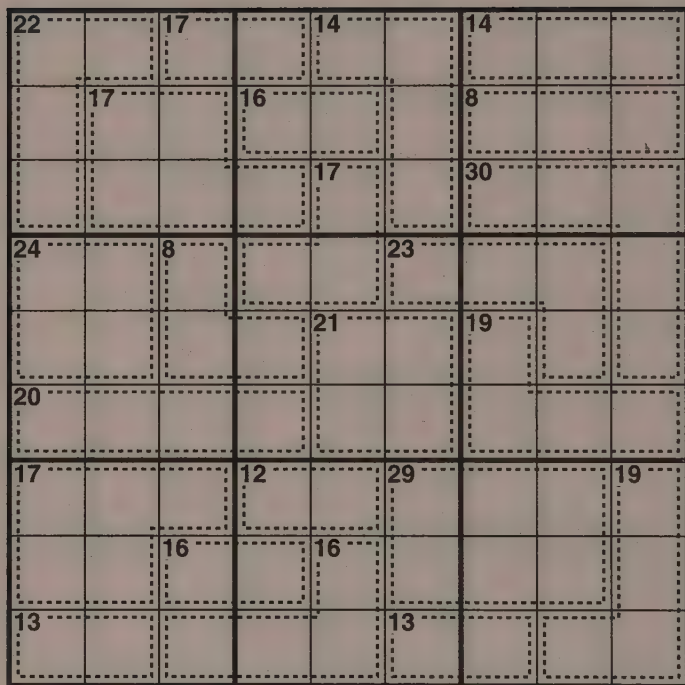
⌚ 1 HOUR 40 MINUTES

TIME TAKEN.....



⌚ 1 HOUR 40 MINUTES

TIME TAKEN.....



⌚ 1 HOUR 40 MINUTES

TIME TAKEN.....

27	7	8	21		20			
					28		6	
		13	9			15	12	
	20		17					
		32		9	21		23	25
	10							
				13				
12	14	13			12	8		
							10	

⌚ 1 HOUR 40 MINUTES

TIME TAKEN.....

21	18		13	16	17		13	
						25		
		16		8				
6					29		14	
19			13					
14		14			23	7	20	
8								18
27		19		6	9		12	

⌚ 1 HOUR 50 MINUTES

TIME TAKEN.....

21		22	22	9			13	
					18		18	17
		20		18				
15					9			
			22			8		
7		14		16	22			8
17	13				10			
			13	10			24	
				19				

⌚ 1 HOUR 50 MINUTES

TIME TAKEN.....

9		23	5	22		17	10	12
4								
9			23	10	9			13
17	22					12		
			12	20				
		16		6	17	14	20	
							13	
9			13	14	10			
17						7		

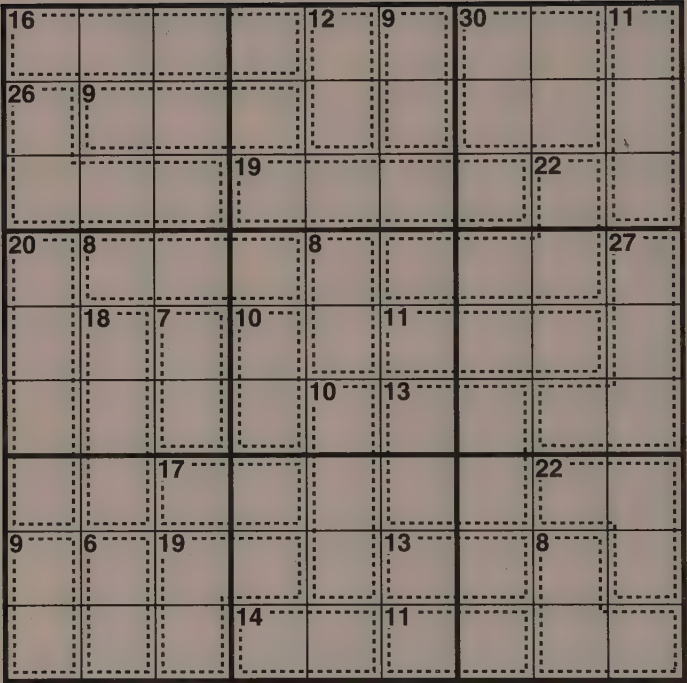
⌚ 1 HOUR 50 MINUTES

TIME TAKEN.....

10		26			3		23	
		30			20			
			13	21		16		
22							9	
		16				17	9	10
28	12	14	12		12			
						21		15
			17					
		13		16				

⌚ 1 HOUR 50 MINUTES

TIME TAKEN.....



⌚ 1 HOUR 50 MINUTES

TIME TAKEN.....

9	21		3		24		8	
				7	20		14	
26		11				12	17	
20		22		17			10	
		12	12		8	11		
15				30		13		13
					9			
22				4		15		

⌚ 1 HOUR 50 MINUTES

TIME TAKEN.....

7		16		9	15	21		14
8	20							
			20		9		10	
20				16		22		6
		33					11	
8				21	7			
		10					17	
20		26			13	5	8	
							13	

⌚ 1 HOUR 50 MINUTES

TIME TAKEN.....

Extra Deadly

5		15		20		12		10
20						23		
18	11		21	12				
				19			11	
	7				11		28	
	16		9		11			16
19		11	29		4			
						7		
12		6			22			

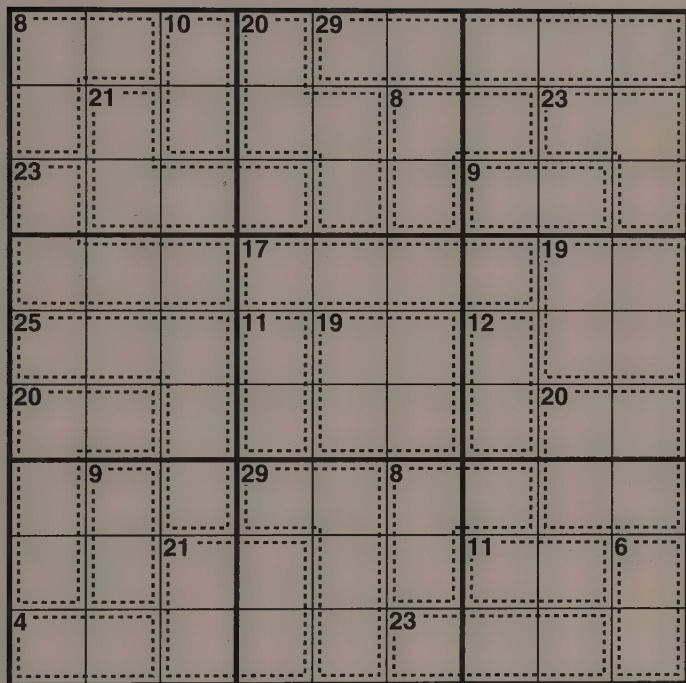
⌚ 1 HOUR 55 MINUTES

TIME TAKEN.....

22	19				6	14		18
		19		18		11		
11	13				8			
		16			4		7	
4		21	11		9		30	
16				12	17		8	
						23		
7	15	7					9	6
		8			16			

⌚ 1 HOUR 55 MINUTES

TIME TAKEN.....



⌚ 1 HOUR 55 MINUTES

TIME TAKEN.....

12	16			20	16			22
					9			
29		9	26			13	7	
9							16	
		22	7		10			
			19			25		
26					17		19	
16			7			23		
		10						

 1 HOUR 55 MINUTES

TIME TAKEN.....

5	12	23		22	21		19	
		15						
27	3		20		12	10	10	
		20		8			23	
			20			11		20
4		6		11			19	
9	14	20						
				21				

⌚ 1 HOUR 55 MINUTES

TIME TAKEN.....

12		22	4		32			
11			19	9	23			
						7		
12				16			29	
8	17				24	23		
		23					10	
10	21			16				
		13			7	10	8	4
		15						

⌚ 2 HOURS

TIME TAKEN.....

18		9	18			23		13
				10	19			
23			23			11		12
	21					8		
		7	23		8			
20				19		10	25	
					20			
9	6	17						14
				19				

⌚ 2 HOURS

TIME TAKEN.....

18					16		22	
11		25		16	14			
21						7	21	
	23		33					
		12			12	21	8	
							19	7
9			9		22			
24		13						22

⌚ 2 HOURS

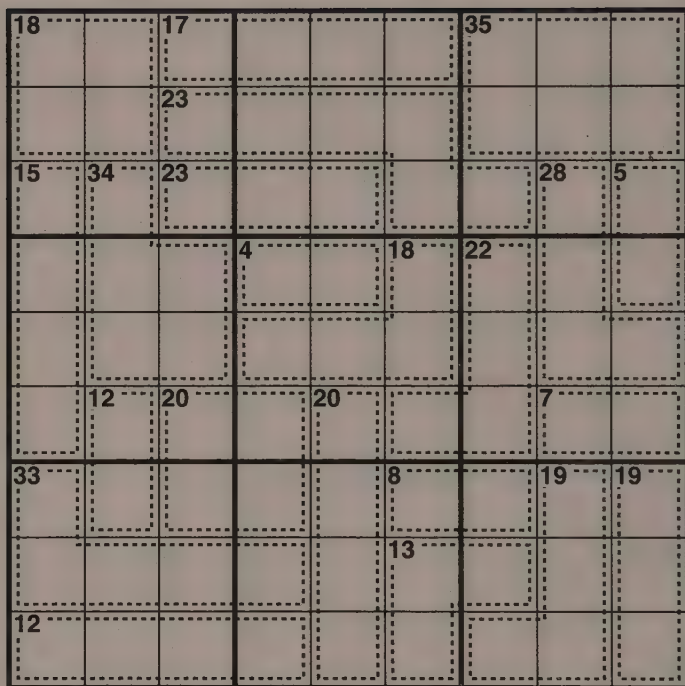
TIME TAKEN.....

25			10		26			
24	18				20			10
				20		8		
		14					17	
15				21				19
	12		16		14	17	9	
24			22					
10		18					16	

⌚ 2 HOURS

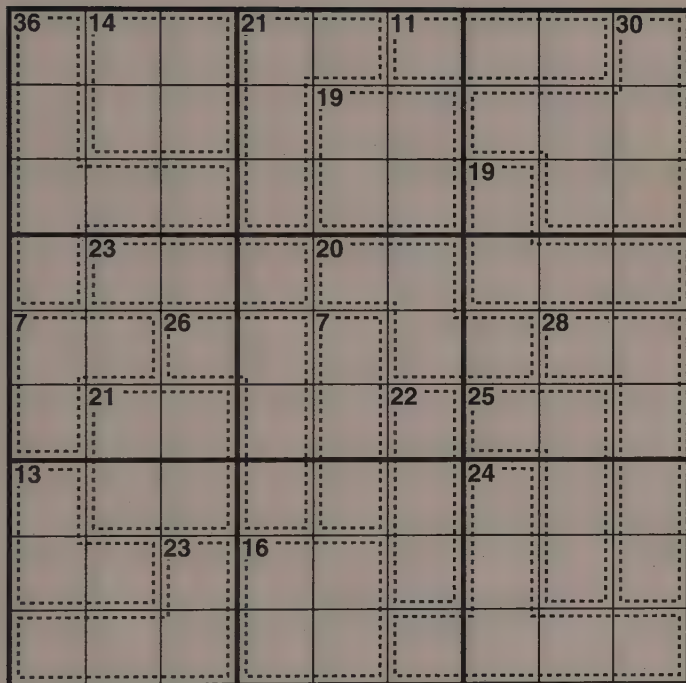
TIME TAKEN.....

Extra Deadly



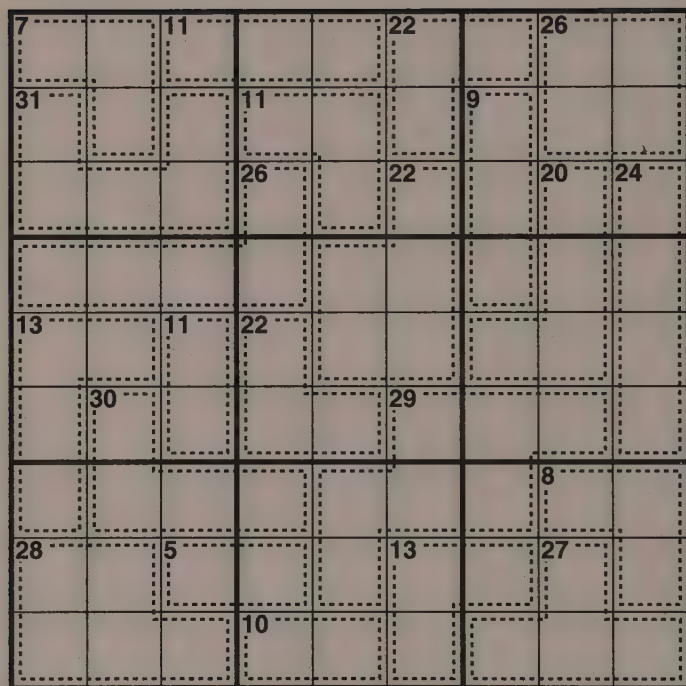
⌚ 2 HOURS 15 MINUTES

TIME TAKEN.....



⌚ 2 HOURS 15 MINUTES

TIME TAKEN.....



⌚ 2 HOURS 15 MINUTES

TIME TAKEN.....

11	21		19	11			14	
					17		7	
20		8			11			17
		20		7	19			
15		15				12		
16			16			15	18	
		7		22			9	
22								
			6		30			

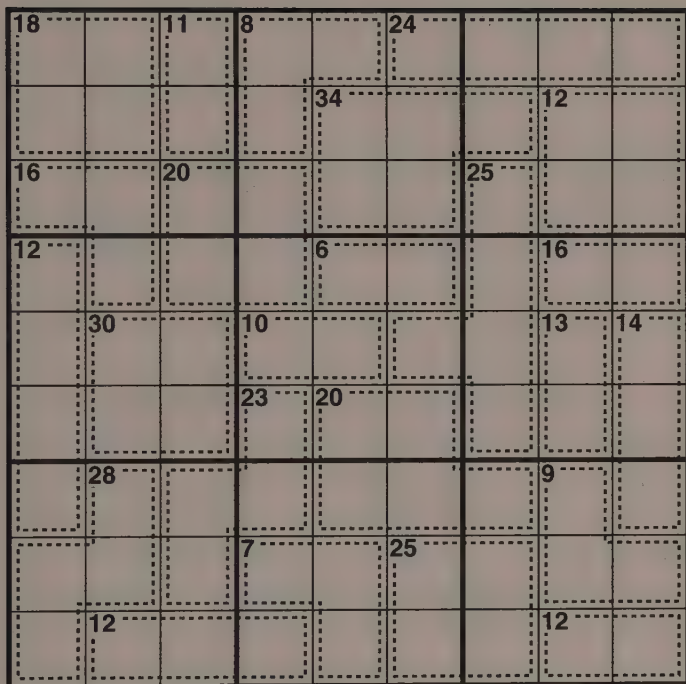
⌚ 2 HOURS 15 MINUTES

TIME TAKEN.....

23	8	17		22	11	7		30
	16			20		8		8
		25	16					
				14			24	
19				10				15
14					19			
	19	9		30			21	

⌚ 2 HOURS 25 MINUTES

TIME TAKEN.....



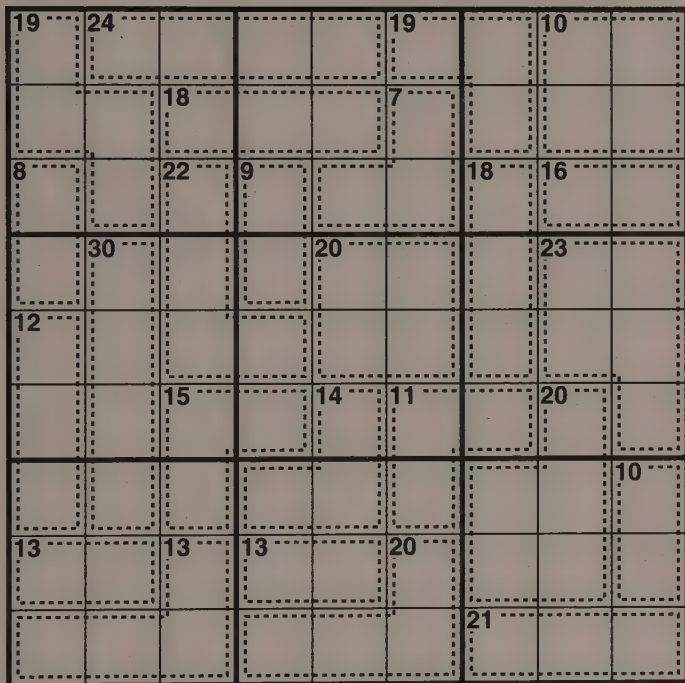
⌚ 2 HOURS 25 MINUTES

TIME TAKEN.....

23		22	27		19	3		25
25	20		13		20	8		
						9	20	
		25						
		9	12			21	19	
				21				
20							8	
20				8			8	

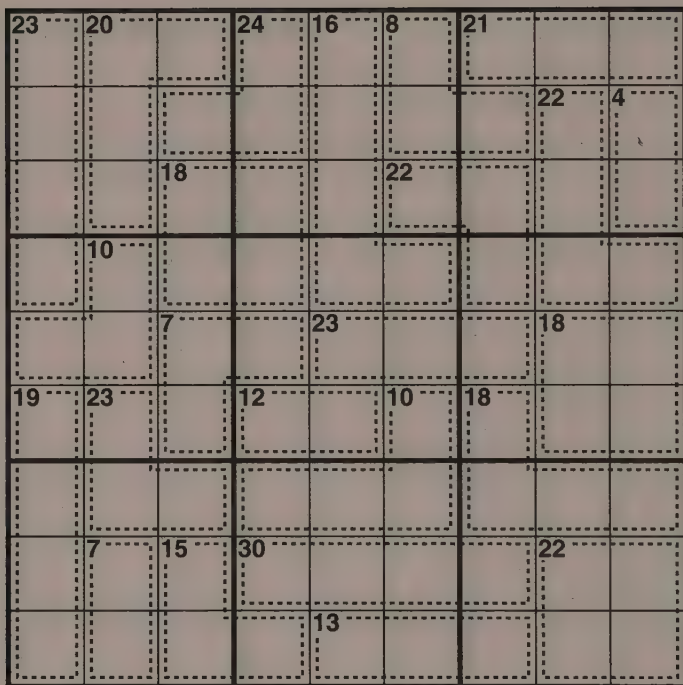
⌚ 2 HOURS 30 MINUTES

TIME TAKEN.....



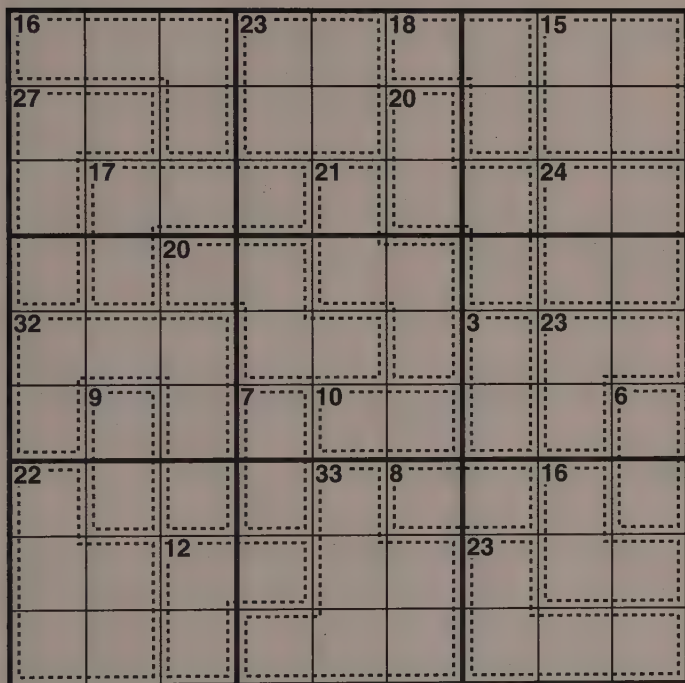
⌚ 2 HOURS 30 MINUTES

TIME TAKEN.....



⌚ 2 HOURS 30 MINUTES

TIME TAKEN.....



⌚ 2 HOURS 30 MINUTES

TIME TAKEN.....

24		27			36			
				20		7		7
12		22		31				
30							19	
19					22		13	15
20			24					
				20				
16					21			

⌚ 2 HOURS 30 MINUTES

TIME TAKEN.....

Cage Reference

Cage combination reference

These are the various combinations of numbers for every possible total covering each of the cage sizes. This should save you time when checking the possibilities.

Cages with 2 squares:

Total	Combinations
3	12
4	13
5	14 23
6	15 24
7	16 25 34
8	17 26 35
9	18 27 36 45
10	19 28 37 46
11	29 38 47 56
12	39 48 57
13	49 58 67
14	59 68
15	69 78
16	79
17	89

Cages with 3 squares:

Total	Combinations
6	123
7	124
8	125 134
9	126 135 234
10	127 136 145 235
11	128 137 146 236 245
12	129 138 147 156 237 246 345
13	139 148 157 238 247 256 346
14	149 158 167 239 248 257 347 356
15	159 168 249 258 267 348 357 456
16	169 178 259 268 349 358 367 457
17	179 269 278 359 368 458 467
18	189 279 369 378 459 468 567
19	289 379 469 478 568
20	389 479 569 578
21	489 579 678
22	589 679
23	689
24	789

Cages with 4 squares:

Total	Combinations
10	1234
11	1235
12	1236 1245
13	1237 1246 1345
14	1238 1247 1256 1346 2345
15	1239 1248 1257 1347 1356 2346
16	1249 1258 1267 1348 1357 1456 2347 2356
17	1259 1268 1349 1358 1367 1457 2348 2357 2456
18	1269 1278 1359 1368 1458 1467 2349 2358 2367 2457 3456
19	1279 1369 1378 1459 1468 1567 2359 2368 2458 2467 3457
20	1289 1379 1469 1478 1568 2369 2378 2459 2468 2567 3458 3467
21	1389 1479 1569 1578 2379 2469 2478 2568 3459 3468 3567
22	1489 1579 1678 2389 2479 2569 2578 3469 3478 3568 4567
23	1589 1679 2489 2579 2678 3479 3569 3578 4568
24	1689 2589 2679 3489 3579 3678 4569 4578
25	1789 2689 3589 3679 4579 4678
26	2789 3689 4589 4679 5678
27	3789 4689 5679
28	4789 5689
29	5789
30	6789

Cages with 5 squares:

Total	Combinations
15	12345
16	12346
17	12347 12356
18	12348 12357 12456
19	12349 12358 12367 12457 13456
20	12359 12368 12458 12467 13457 23456
21	12369 12378 12459 12468 12567 13458 13467 23457
22	12379 12469 12478 12568 13459 13468 13567 23458 23467
23	12389 12479 12569 12578 13469 13478 13568 14567 23459 23468 23567
24	12489 12579 12678 13479 13569 13578 14568 23469 23478 23568 24567
25	12589 12679 13489 13579 13678 14569 14578 23479 23569 23578 24568 34567
26	12689 13589 13679 14579 14678 23489 23579 23678 24569 24578 34568
27	12789 13689 14589 14679 15678 23589 23679 24579 24678 34569 34578
28	13789 14689 15679 23689 24589 24679 25678 34579 34678
29	14789 15689 23789 24689 25679 34589 34679 35678
30	15789 24789 25689 34689 35679 45678
31	16789 25789 34789 35689 45679
32	26789 35789 45689
33	36789 45789
34	46789
35	56789

Cages with 6 squares:

Total	Combinations
21	123456
22	123457
23	123458 123467
24	123459 123468 123567
25	123469 123478 123568 124567
26	123479 123569 123578 124568 134567
27	123489 123579 123678 124569 124578 134568 234567
28	123589 123679 124579 124678 134569 134578 234568
29	123689 124589 124679 125678 134579 134678 234569 234578
30	123789 124689 125679 134589 134679 135678 234579 234678
31	124789 125689 134689 135679 145678 234589 234679 235678
32	125789 134789 135689 145679 234689 235679 245678
33	126789 135789 145689 234789 235689 245679 345678
34	136789 145789 235789 245689 345679
35	146789 236789 245789 345689
36	156789 246789 345789
37	256789 346789
38	356789
39	456789

Cages with 7 squares:

Total	Combinations
28	1234567
29	1234568
30	1234569 1234578
31	1234579 1234678
32	1234589 1234679 1235678
33	1234689 1235679 1245678
34	1234789 1235689 1245679 1345678
35	1235789 1245689 1345679 2345678
36	1236789 1245789 1345689 2345679
37	1246789 1345789 2345689
38	1256789 1346789 2345789
39	1356789 2346789
40	1456789 2356789
41	2456789
42	3456789

Cages with 8 squares:

Total	Combinations		
36	12345678	41	12356789
37	12345679	42	12456789
38	12345689	43	13456789
39	12345789	44	23456789
40	12346789		

Solutions

1

3	2	6	5	7	9	8	1	4
4	9	8	6	1	3	7	5	2
5	1	7	8	2	4	3	6	9
1	5	4	2	8	7	6	9	3
6	8	3	4	9	5	1	2	7
9	7	2	3	6	1	5	4	8
7	6	5	9	4	8	2	3	1
8	3	9	1	5	2	4	7	6
2	4	1	7	3	6	9	8	5

2

9	3	2	6	5	1	4	8	7
4	7	1	2	8	9	6	3	5
5	8	6	4	3	7	1	9	2
7	5	3	8	1	2	9	4	6
2	1	4	3	9	6	7	5	8
8	6	9	7	4	5	3	2	1
1	9	8	5	6	3	2	7	4
3	4	7	1	2	8	5	6	9
6	2	5	9	7	4	8	1	3

3

9	5	7	8	6	3	2	4	1
4	6	2	5	9	1	7	8	3
1	3	8	2	4	7	6	5	9
5	2	1	3	8	6	9	7	4
7	8	4	9	5	2	3	1	6
6	9	3	7	1	4	8	2	5
3	4	9	1	7	8	5	6	2
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4

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3	6	8	7	9	1	5	4	2
9	4	2	3	8	5	7	6	1
1	5	7	6	4	2	8	3	9
5	3	4	9	1	7	6	2	8
7	2	6	8	5	3	1	9	4
8	9	1	4	2	6	3	7	5

5

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2	9	7	6	1	8	4	3	5
5	3	4	2	7	9	6	1	8
4	7	5	1	6	2	3	8	9
6	1	9	4	8	3	5	7	2
3	8	2	9	5	7	1	6	4
9	5	6	7	2	1	8	4	3
7	4	3	8	9	6	2	5	1
1	2	8	3	4	5	7	9	6

6

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9	5	3	8	7	4	1	2	6
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7

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6	8	9	1	5	4	7	2	3
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9	4	6	8	1	7	3	5	2
8	2	7	5	4	3	6	1	9
3	7	8	2	9	1	4	6	5
5	6	2	4	3	8	9	7	1
4	9	1	6	7	5	2	3	8

8

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6	1	9	3	7	5	2	4	8
7	3	8	9	1	6	4	2	5
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1	4	5	7	2	8	9	3	6
9	5	4	1	3	7	8	6	2
8	2	1	6	4	9	5	7	3
3	6	7	5	8	2	1	9	4

9

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5	8	9	4	1	2	7	3	6
2	7	1	3	6	9	8	4	5
4	9	2	6	3	5	1	7	8
7	1	5	8	9	4	3	6	2
3	6	8	1	2	7	5	9	4
8	2	6	9	7	3	4	5	1
9	5	7	2	4	1	6	8	3
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10

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5	4	6	3	8	1	2	7	9
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9	2	8	1	6	5	7	4	3
4	5	7	2	9	3	6	8	1
6	9	5	8	1	4	3	2	7
8	1	3	7	2	9	4	5	6
2	7	4	5	3	6	9	1	8

11

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2	8	7	1	6	5	4	9	3
1	9	6	4	3	2	7	5	8
8	5	9	6	1	3	2	7	4
7	2	1	8	5	4	3	6	9
6	4	3	2	9	7	1	8	5
9	6	5	3	4	1	8	2	7
3	1	2	9	7	8	5	4	6
4	7	8	5	2	6	9	3	1

12

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1	8	2	5	3	4	9	6	7
3	4	8	9	1	6	7	2	5
7	2	6	4	8	5	1	3	9
9	5	1	3	7	2	6	8	4
6	7	5	2	9	8	3	4	1
8	3	9	6	4	1	5	7	2
2	1	4	7	5	3	8	9	6

13

8	7	6	5	1	9	2	4	3
3	2	9	6	7	4	5	8	1
5	4	1	8	2	3	7	9	6
4	8	7	3	6	2	9	1	5
6	1	5	7	9	8	4	3	2
2	9	3	1	4	5	8	6	7
1	3	4	9	5	7	6	2	8
7	6	2	4	8	1	3	5	9
9	5	8	2	3	6	1	7	4

14

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5	2	9	8	6	3	4	7	1
7	6	4	1	2	9	8	5	3
2	7	1	3	5	4	6	8	9
6	3	8	9	1	7	2	4	5
9	4	5	6	8	2	1	3	7
1	8	2	5	3	6	7	9	4
3	9	7	2	4	8	5	1	6
4	5	6	7	9	1	3	2	8

15

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3	2	9	1	7	5	6	8	4
8	9	4	6	3	7	2	1	5
7	3	2	4	5	1	8	9	6
1	5	6	2	8	9	7	4	3
9	1	5	7	4	6	3	2	8
2	4	7	3	1	8	5	6	9
6	8	3	5	9	2	4	7	1

16

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3	9	1	8	7	6	2	5	4
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8	4	9	2	5	7	3	6	1
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17

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3	7	8	5	2	6	1	4	9
4	3	9	2	7	5	8	1	6
1	8	7	6	9	4	5	3	2
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18

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9	5	2	8	7	1	3	6	4
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5	9	3	7	1	2	6	4	8

19

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1	6	2	3	8	5	9	7	4

20

4	3	6	1	5	8	7	2	9
1	5	9	4	7	2	3	8	6
7	8	2	3	9	6	1	5	4
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21

7	8	9	6	3	2	4	1	5
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9	4	5	2	1	3	6	8	7
3	7	1	8	9	6	2	5	4
8	2	6	4	7	5	9	3	1

22

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7	4	6	3	8	5	2	9	1
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23

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8	9	6	3	1	7	2	4	5
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1	6	8	2	7	3	9	5	4
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24

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5	4	8	3	6	7	9	1	2
3	6	9	2	1	5	8	7	4

25

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26

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9	4	2	6	3	5	8	1	7
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3	1	9	7	2	4	5	8	6
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27

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28

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6	4	1	8	9	7	2	5	3

29

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30

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31

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32

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33

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34

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35

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77

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79

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80

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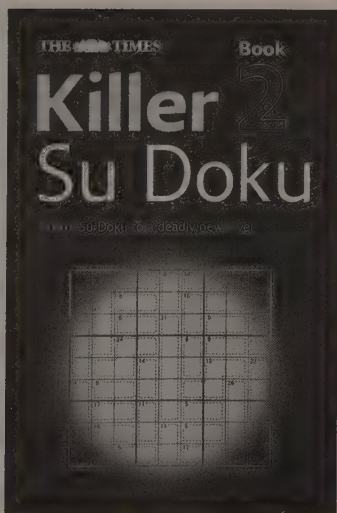
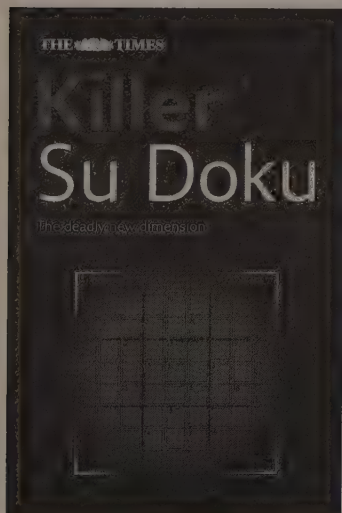
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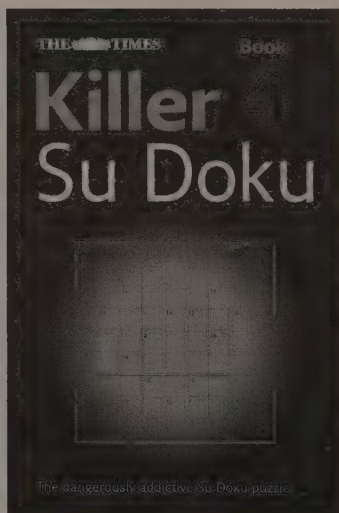
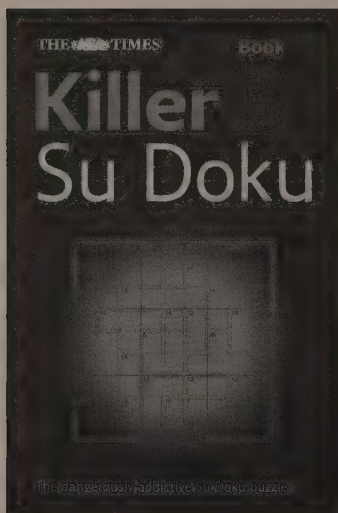


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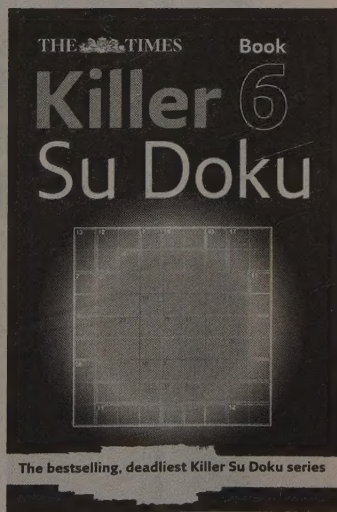
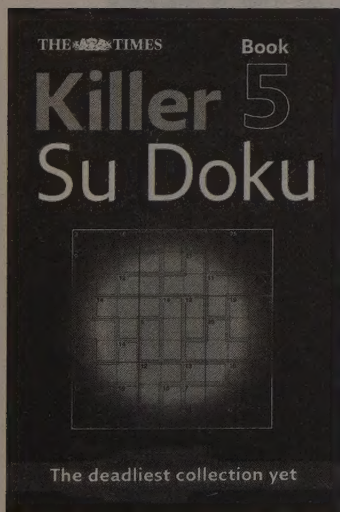


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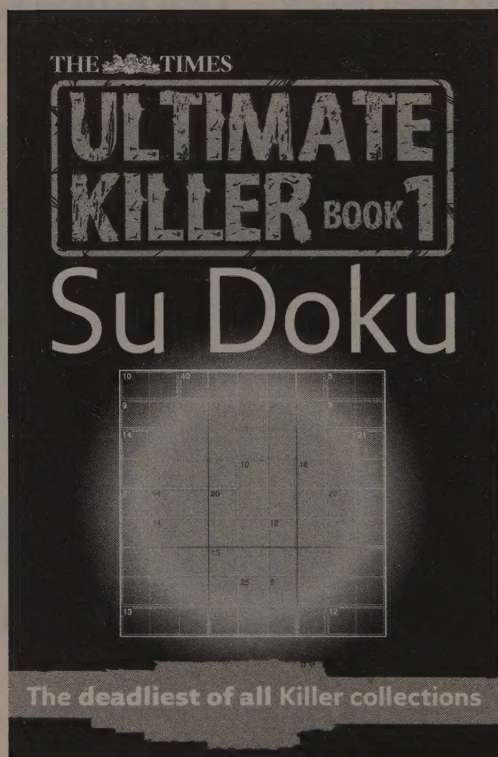


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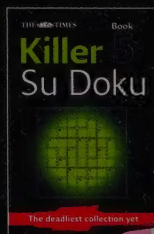
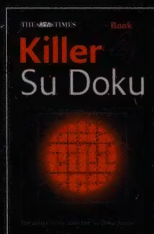
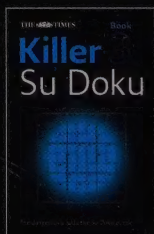
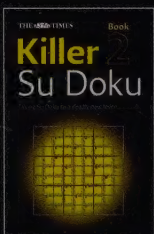
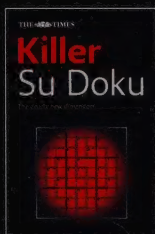


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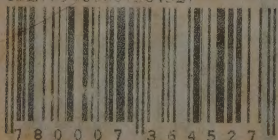
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